



RV College of
Engineering®



IV Semester
AEROSPACE PROPULSION
AS343AI

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Course Curriculum

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Semester : IV					
AEROSPACE PROPULSION					
Category: PROFESSIONAL CORE COURSE (Theory & Practice)					
Course Code	:	AS343AI		CIE	: 100 +50 Marks
Credits: L:T:P	:	3:0:1		SEE	: 100 +50 Marks
Total Hours	:	45L+28P		SEE Duration	: 3.00 +3.00 Hours

Unit-I	10 Hrs
Fundamentals of Aerospace propulsion: Introduction, Illustration of working of gas turbine engine, Working and characteristics of Turbojet, Turboprop, Turbofan, Ramjet & Scramjet Engines, Spooling of Jet Engines, Thrust Augmentation System: Afterburner and Water Injection.	
Unit – II	10 Hrs
Jet Engine Performance: Jet Engine Thrust equation (simple derivation), Factors affecting thrust, Airbreathing Engine Performance Parameters: Specific Thrust, Thrust Power, Specific Impulse, TSFC, Propulsive Efficiency, Thermal Efficiency, Numericals on Jet Propulsion Cycle Analysis.	
Unit –III	08 Hrs
Aircraft Propulsion Systems: Aircraft Inlets-Working of Subsonic & Supersonic Inlets, Compressors: Operation of Centrifugal & Axial Flow Compressors, Combustion Chambers: Principle of operation, Classification of Combustion Chambers, Turbines: Types of turbines-Operating Principle, Nozzles: Flow through Convergent and Convergent-Divergent Nozzles (Without Derivations and Numericals).	



Unit –IV		07 Hrs
Chemical Rocket Propulsion: Solid Rocket propellants: Working, Properties, Types of Propellants, Propellant Burn Rate, Thrust profiles, Attitude Controls with Solid Propellants, Pyrogen and Pyrotechnique Igniters, Ignition and Thrust Extinction of Solid Propellants. Liquid Rocket propellants: Working, Propellant Types, Properties, Propellant feed systems: Pump and Gas Pressure Feed systems, Injectors and Atomizers.		
Unit –V		10 Hrs
Advanced Propulsion Systems: Electrothermal Thrusters: Arc-jet Thrusters, Ion Propulsion, Plasma Thrusters: Hall Effect Thrusters, Electric Power Generation: Solar Cells, Solar Generators, Radioactive Thermal Generators, Nuclear Fission Power Generators. Rocket Performance: Rocket equation, Performance Parameters: Thrust, Total Impulse, Specific Impulse, Specific propellant consumption, Effective Exhaust Velocity, Characteristic Velocity, Mass Ratio, Propellant Mass Fraction, Impulse to weight ratio, Thrust to weight ratio, Energy and Efficiencies, Numerical examples.		
Reference Books		
1	Gas Turbines, V Ganesan, 3 rd Edition, 2017, McGraw Hill Education, ISBN-10: 0070681929	
2	Gas Turbine Propulsion, D P Mishra, 2 nd Edition, M V Learning, 2015, ISBN: 978-81-309-27527	
3	Elements of Propulsion: Gas Turbines and Rockets, Jack D Mattingly, 5 th Edition, 2006, American Institute of Aeronautics and Astronautics (AIAA), ISBN: 1563477793.	
4	Rocket Propulsion Elements, Sutton G P, 8 th Edition, 2010, John Wiley, New York, ISBN:9781118174203	
5	Understanding Aerospace Chemical Propulsion, H S Mukunda, 1 st Edition, 2017, I K International Publishing House, ISBN: 978-93-85909-42-9	
6	Rocket and Spacecraft Propulsion: Principles, Practices and Developments, Martin J L Turner, 3 rd Edition, 2009, Praxis Publishing Ltd, Chichester, UK, ISBN 978-3-540-69202-7	



LABORATORY EXPERIMENTS

1. Performance analysis of a micro gas turbine/jet propulsion system
2. Determination of Performance characteristics of a fixed pitch aircraft propeller
3. Determination of Performance characteristics of a variable pitch aircraft propeller-Open Experiment
4. Measurement of burning velocity of a pre-mixed flame in a gas turbine combustion chamber
5. Determine the pressure and velocity variation of an exhaust gas flowing out of a convergent nozzle
6. Determination of pressure and velocity variation of a supersonic exhaust jet flowing out of a convergent-divergent nozzle
7. Study of pressure distribution across a turbine cascade
8. Study of flow through an axial cascade turbine blade row
9. Preparation of Solid Propellant Rocket Fuel- Open Experiment
10. Evaluation of Burning Characteristics of Solid Propellant Fuel- Open Experiment



RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY & PRACTICE)		
#	COMPONENTS	MARKS
1.	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & Each Quiz will be evaluated for 10 Marks adding up to 20 Marks. THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.	20
2.	TESTS: Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.	40
3.	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Phase I (20) & Phase II (20) ADDING UPTO 40 MARKS.	40
4.	LAB: Conduction of laboratory exercises, lab report, observation, and analysis (20 Marks),lab test (10 Marks) and Innovative Experiment/ Concept Design and Implementation (20 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE 50 MARKS	50
MAXIMUM MARKS FOR THE CIE (THEORY & PRACTICE)		150

RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q. NO	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	30
3	Viva	10
TOTAL		50



RUBRIC FOR SEMESTER END EXAMINATION (THEORY)

Q. NO	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	20
PART B (Maximum of THREE Sub-divisions only)		
2	Unit 1: (Compulsory)	16
3 & 4	Unit 2: Question 3 or 4	16
5 & 6	Unit 3: Question 5 or 6	16
7 & 8	Unit 4: Question 7 or 8	16
9 & 10	Unit 5: Question 9 or 10	16
TOTAL		100



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History of Aviation

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- *Aircraft-Greatest inventions*
- *Evolution-Desire to fly*
- *Evident in many Mythologies*
 - *Greek, Persian, Hindu etc*



Greek Mythology



Flying Carpets- Persian Mythology



Pushpaka Vimana- Hindu Mythology

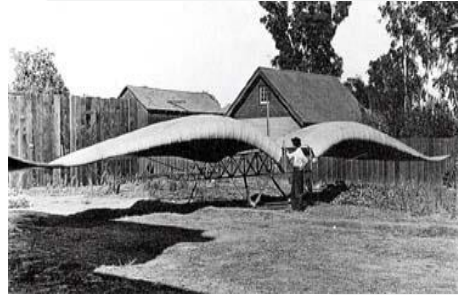
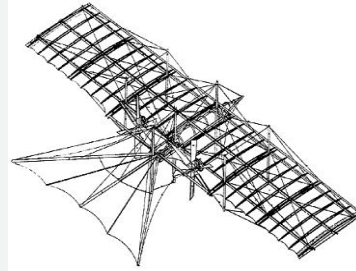


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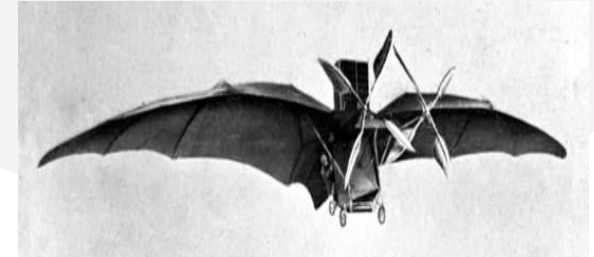
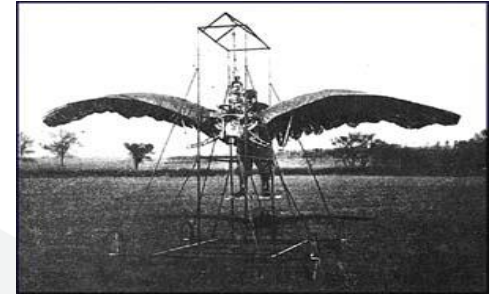
Early Adventures

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- *Gazing of Birds and Insects*
- *Studying their Flying characteristics*
- *Incorporation of Drag force in the design*
- *Plagued with Repeated failures*
- *Fair Knowledge- Aerodynamics*
- ***Failures & Failures & Failures.....***



Early Airplane Designs





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Early Adventures Contd...

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VIDEO ON INTRODUCTION TO PROPULSION



- *Failures – Due to insufficient source of power*

Era of Wright Brothers

- Embarked on methodological approach
- Studied many gliders
- Plagued by failures
- Built Wind Tunnel-*Studied Aerodynamics of wings*
- Calculation of Lift and Drag
- Incorporation of Propellers, IC Engine, Yaw and Roll control systems
- ***December 04, 1903 Langley Aerodrome fails for the second time***
- ***December 13, 1903 Wilber Wright stalls and damages the Flyer***
- ***December 17, 1903, in the repaired Flyer***
 - Orville Wright, 1st flight, 112 feet
 - Wilber Wright, 2nd flight, 175 feet
 - Orville Wright, 3rd flight, 200 feet
 - Wilber Wright, 4th flight, 854 feet
 - Then, **5th flight**, flew on its own in a gust of wind and was **wrecked**
- Total flying, Dec 17th – **A minute and a half, about 1300+ feet**



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Video On Wright Flyer

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Old Movie Collector

Wright Brothers
First Flight
1903

IWM



Reciprocating Engines-Otto Cycle

The Various thermodynamic processes in a Piston Engine are

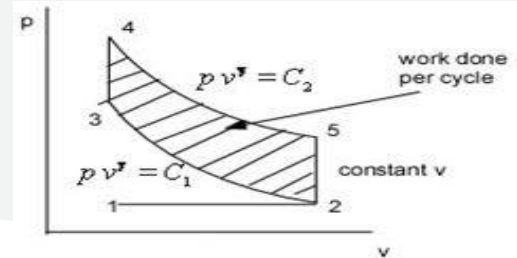
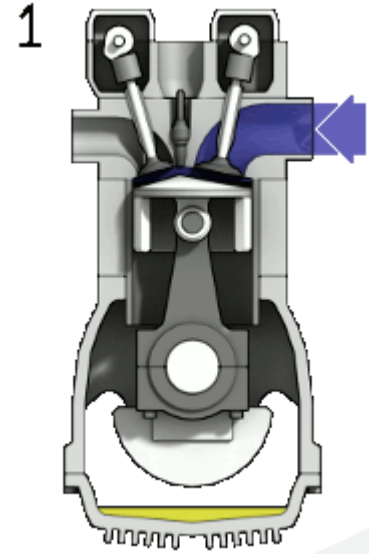
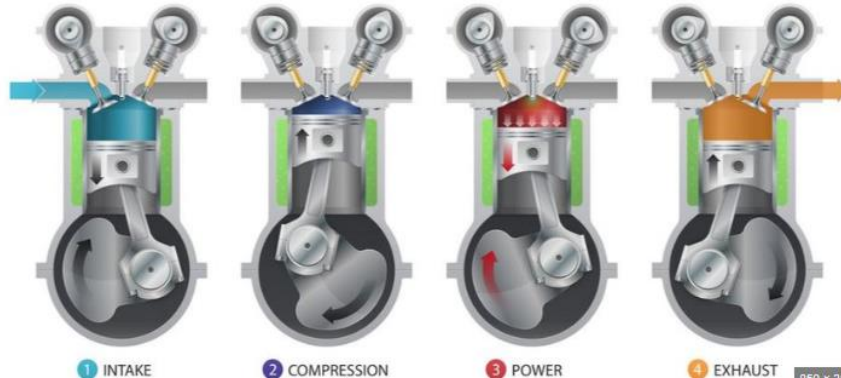
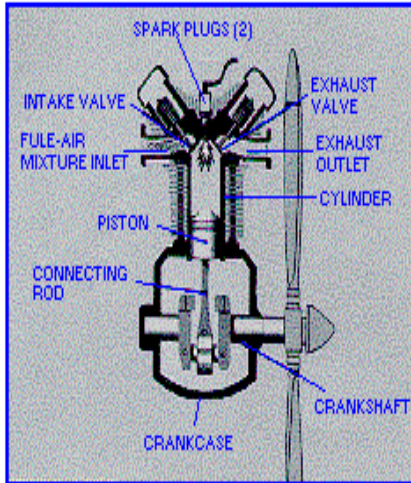
Isentropic Compression

Constant volume heat addition

Isentropic expansion

Constant volume heat rejection

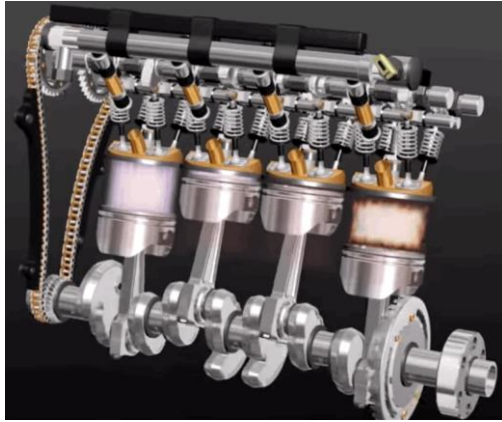
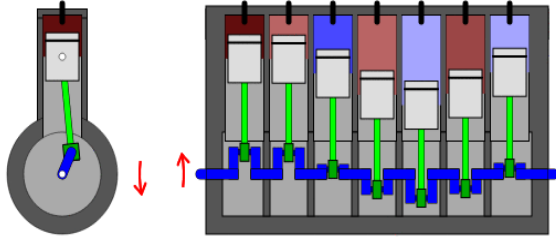
AIRCRAFT PISTON ENGINE-WORKING-4 STROKE



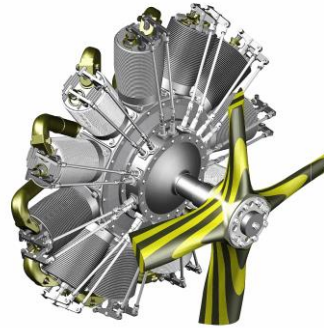
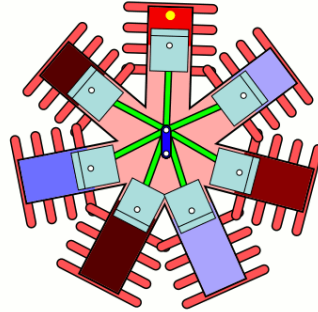


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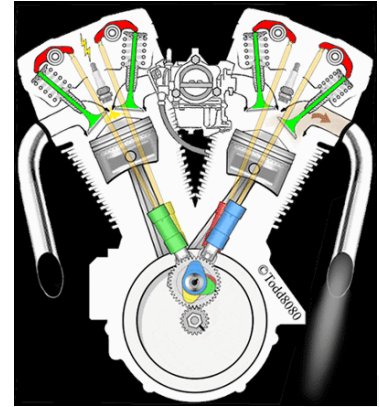
Piston Engine Configurations



In-Line Engine



Rotary Engine

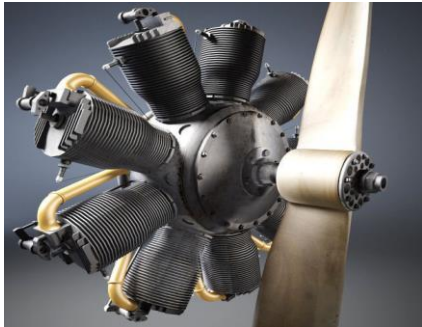
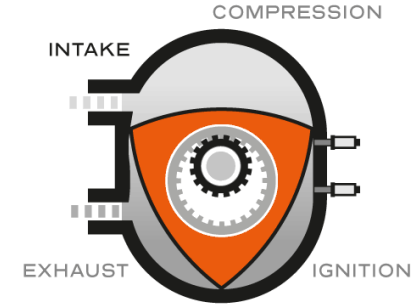


V-Engine



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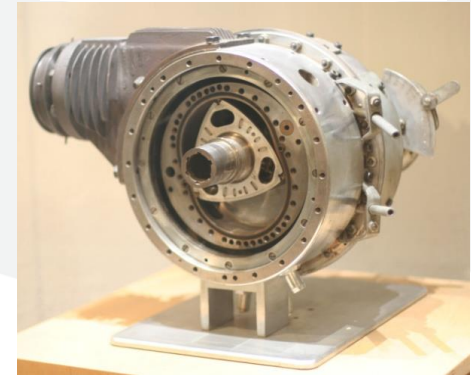
Piston Engine Configurations Contd...



Rotary Engines



Subaru Horizontally-Opposed Boxer Engine



Wankel Engine



Specifications - *Examples*

1) DAIMLER-BENZ DB 602 (1930)

General characteristics

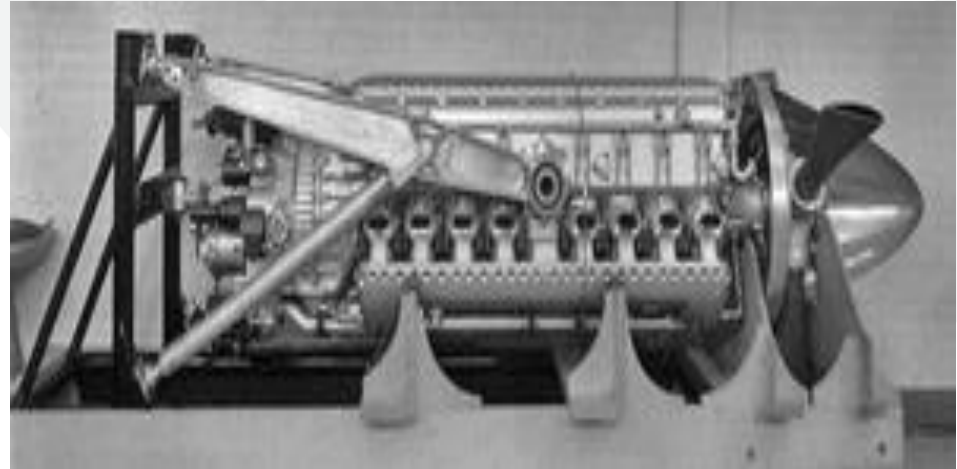
- Type: 16-cylinder liquid-cooled V16 aircraft diesel engine
- Bore: 175 mm
- Stroke: 230 mm
- Displacement: 88.5 L
- Dry weight: 1,976 kg

Components

- Fuel system: Fuel Injection
- Cooling system: Liquid-cooled

Performance

- Power output: 984 kW (1,320 hp)





LYCOMING XR-7755 (1943)

General characteristics

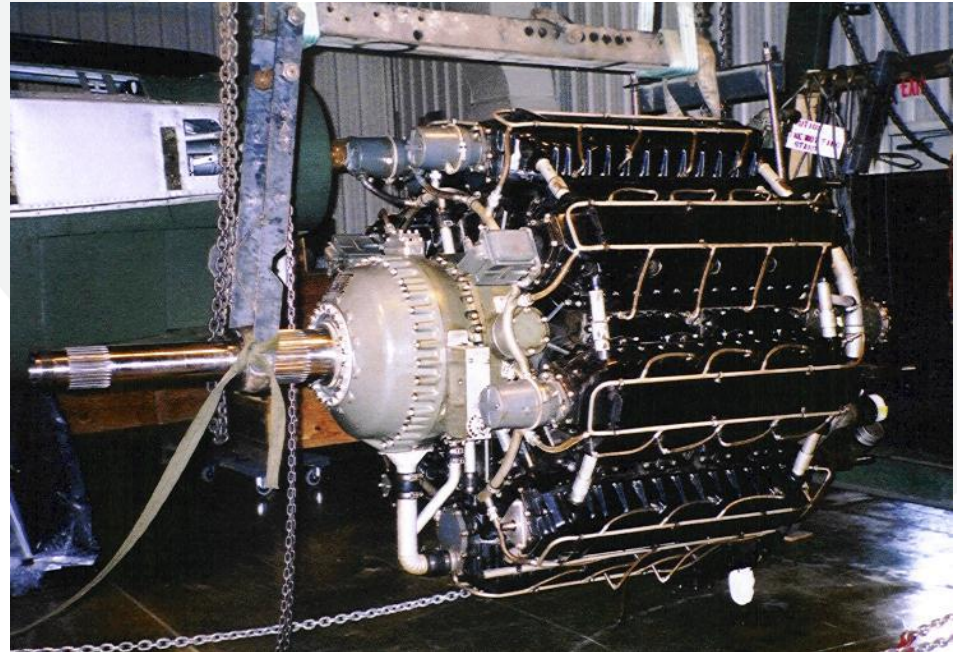
- Type: 36-cylinder turbo supercharged liquid-cooled aircraft piston engine
- Bore: 161.9 mm
- Stroke: 171.4 mm
- Displacement: 127.1 L
- Length: 3,050 mm
- Diameter: 1,525 mm
- Dry weight: 2,745 kg

Performance

- 5,000 hp (3,730 kW) at 2,600 rpm take off
- 4,000 hp (2,985 kW) at 2,300 rpm cruise

Specific fuel consumption

- 0.43 kg/(kW·h) at takeoff power
- 0.29 kg/(kW·h) at 70% power
- 0.22 kg/(kW·h) at minimum cruise power
- Power to weight ratio 1.36 kW/kg





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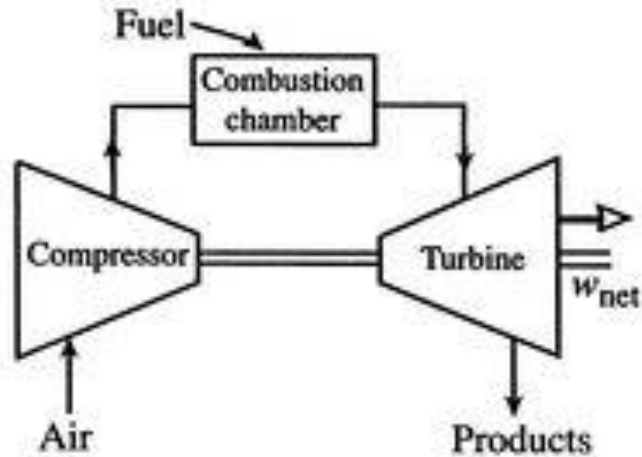
Demerits of IC Engines

- *Inefficient at higher altitude*
- *Increased vibrational problems*
- *Lubrication problems*
- *Cooling problems*
- *Increased frontal area*
- *Increased drag*
- *Increased fuel consumption*
- *Increased complexities in design*
- *Suitable only for small sized aircrafts*

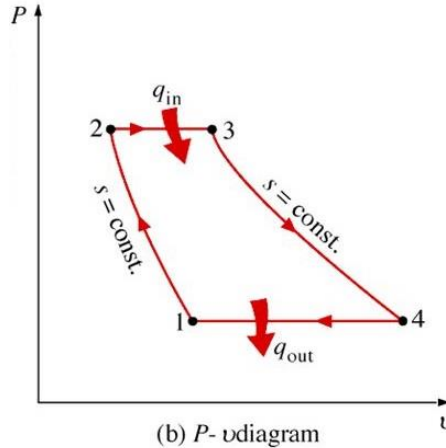


Simple Gas Turbine Engines

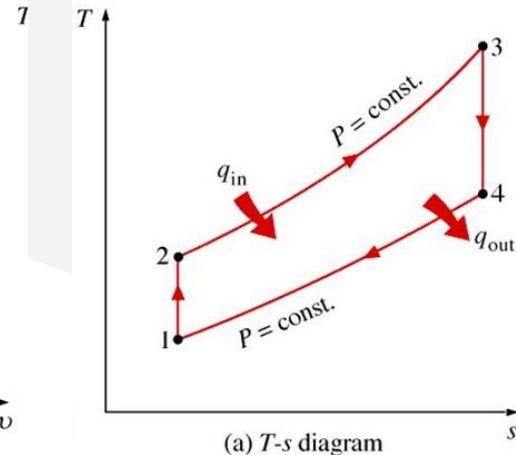
- Works on Brayton cycle-constant pressure cycle
- Consists of *compressor, burner, turbine*
- *Thermodynamic cycle consists of four processes*



- 1-2-isentropic compression
- 2-3-constant pressure heat addition
- 3-4-isentropic expansion
- 4-1-constant pressure heat rejection



(b) P - v diagram



(a) T - s diagram



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Advantages of Gas Turbine Engines

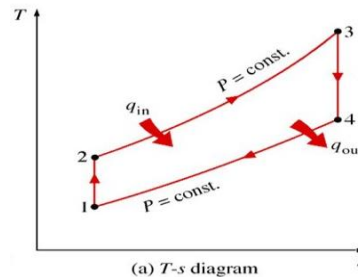
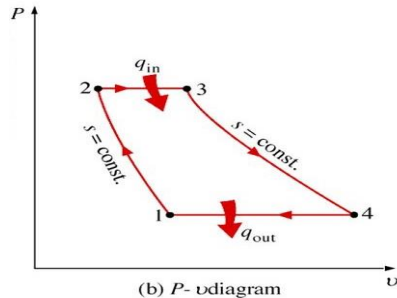
- *Absence of moving components*
- *Absence of rubbing parts in the engine*
- *High reliability*
- *Low lubrication*
- *High power to weight ratio*
- *Ease of maintenance*
- *Operating flexibility*
- *High specific thrust*
- *High thermal efficiencies*
- *Low vibrations*
- *Higher mass flow rates*



RV College of Engineering® Ideal and Real Cycle-Deviations of Brayton Cycle from Ideal Conditions

Assumptions considered in Brayton cycle:

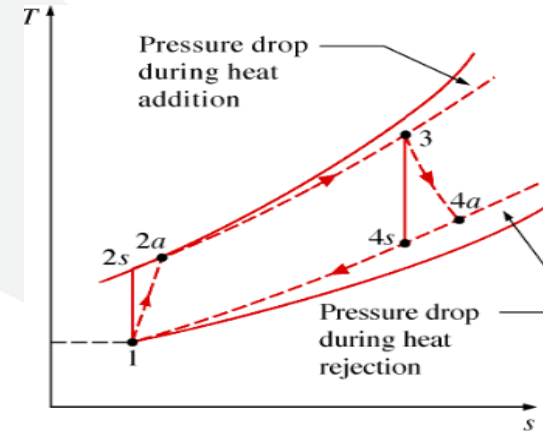
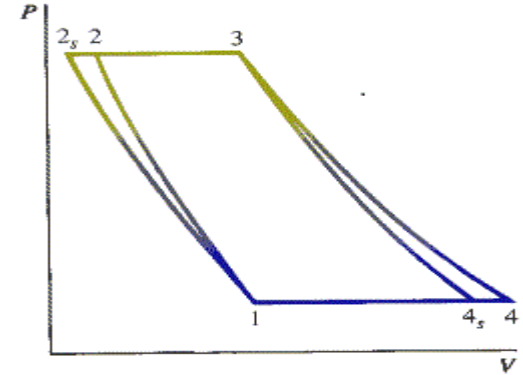
- There are no frictional losses throughout the cycle.
 - There are no irreversibilities occurring in the processes.
 - The processes in the diffuser, compressor, turbine and nozzle are assumed to be isentropic.
 - Air is assumed to behave like Ideal gas with specific heats to remain constant throughout the cycle.
-
- 1-2-isentropic compression – **CONSTANT ENTROPY-REVERSIBLE ADIABATIC**
 - 2-3-constant pressure heat addition -
 - 3-4-isentropic expansion - **CONSTANT ENTROPY-REVERSIBLE ADIABATIC**
 - 4-1-constant pressure heat rejection





Losses in the Compressor

- Various losses occur in a Compressor
- **Frictional forces (Mechanical friction + Fluid Friction)**
 - *Shaft Losses, Bearing Losses etc*
 - *Fluid Friction- Loss of kinetic energy*
 - *Increases the temperature of the working fluid*
 - *Irreversibly converted to internal energy*
 - Leads to **INCREASED ENTROPY-NON-ISENTROPIC FLOW**
- **Turbulence Flows in the compressor**
 - Pressure Losses
- **Formation of Shock Waves in the Compressor**
 - Increases the Temperature of the working fluid
 - Higher Exit temperature
- **Losses lead to power losses**
 - Needs more power than actually required





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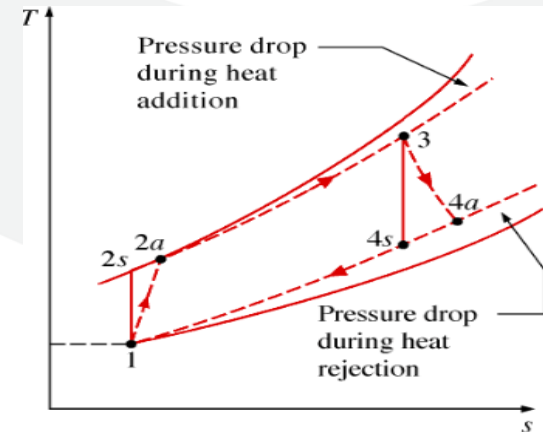
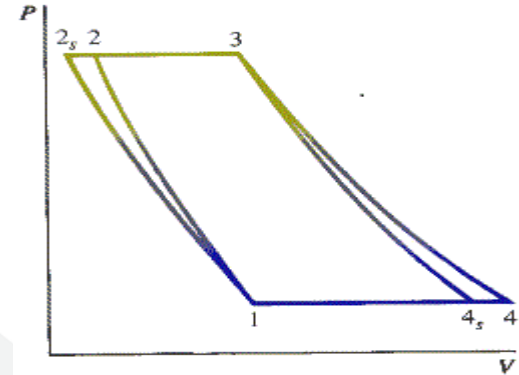
Compressor and Turbine Blades





Losses in the Combustion chamber

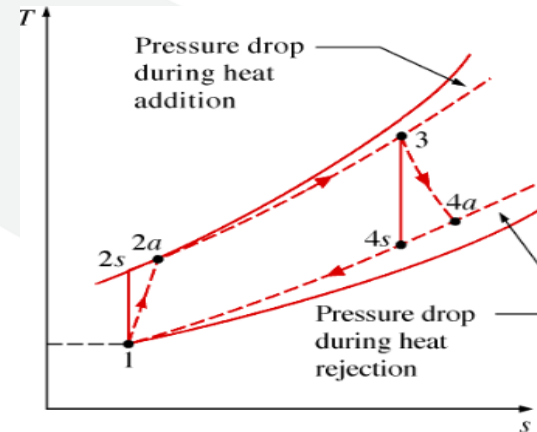
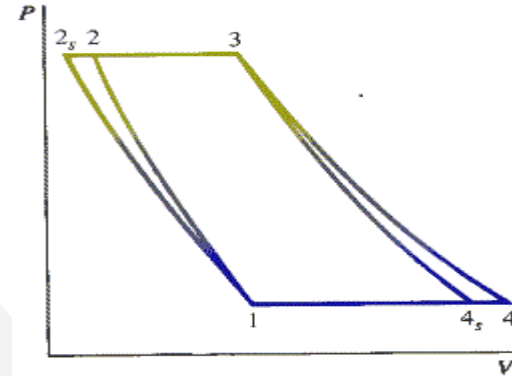
- Admission of High pressure air to the combustion chamber
- Formation of **COMBUSTIBLE** Mixture
 - For mixing of fuel and air
 - Promotion of Turbulence
 - Velocity increases and pressure drops
- **Friction** between the walls and the working fluid
 - Pressure losses
- Hence, **Heat ADDITION-NOT AT CONSTANT PRESSURE**
- Affects the total thrust/velocity of the exhaust gases





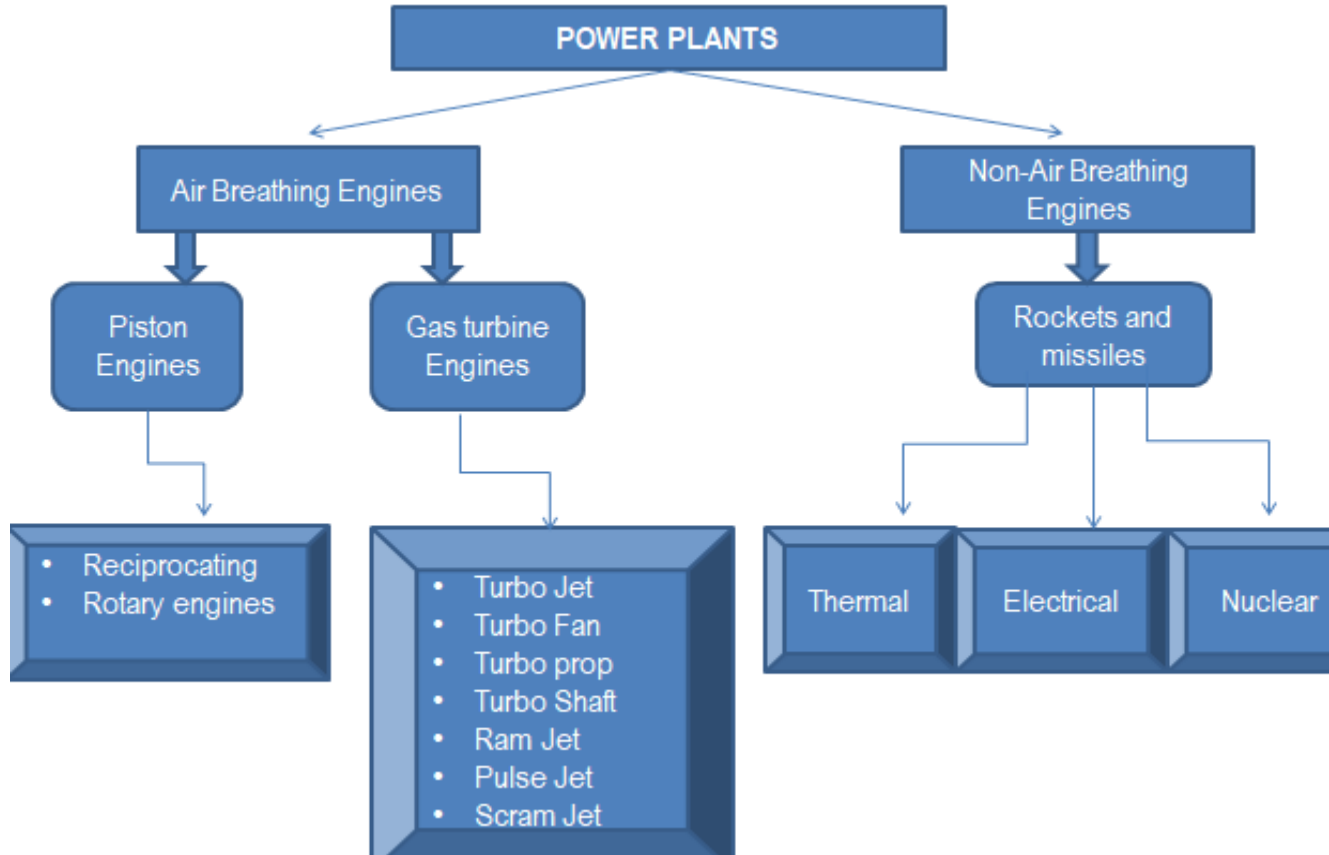
Losses in the Turbine

- Turbine- Expand of exhaust gases
- Losses similar to the compressor
- Turbine-mechanical and fluid friction
 - Deceleration of fluid,
 - Dissipating some kinetic energy into internal energy rise
- Leads to increased entropy
- **Turbulence Flows in the compressor**
 - Pressure Losses
- **Formation of Shock Waves in the Compressor**
 - Increases the Temperature of the working fluid
 - Higher Exit temperature
- **Losses lead to power losses**
 - **GENERATES LESS** power than actually required





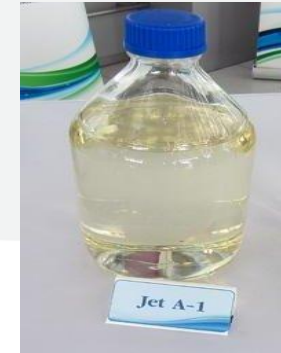
Classification of Power Plants





Commonly used fuels

1. **AVgas (Aviation Gasoline) 100-HL** :High leaded fuel **green** in colour
2. **AVgas 130-LL** :Low Leaded fuel **blue** in colour
3. **AVgas 82- UL** :Unleaded fuel **purple** in colour
4. **AVgas 80/87** :**Red** in colour
5. **Jet A1 or Jet B** : Gas Turbine Engines (Kerosene based)



Fuel Type and Grade	Color of Fuel	Equipment Control Color	Pipe Banding and Marking	Refueler Decal
AVGAS 82UL	Purple			
AVGAS 100	Green			
AVGAS 100LL	Blue			
JET A	Colorless or straw			
JET A-1	Colorless or straw			
JET B	Colorless or straw			



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Jet A1

lower freezing point (-47°C)

suitable for long-haul international flights (overflying polar routes)

static dissipater additives

Common for most airlines

Jet A

freezing point (-40°C)

Jet B

70% gasoline and 30% kerosene

low freezing point of -60°C ,

suitable for extremely cold countries such as Alaska and Canada

TS-1

jet fuel is common in Russia

freezing point of -50°C

Military jet Fuels

JP-1

JP-2

JP-3

JP-4

JP-5

JP-6

JP-7

JP-8

JP-9

JPTS

Zip fuel

Syntroleum





COMPONENTS OF A GAS TURBINE PROPULSION SYSTEM

Most Essential Parts of any Gas Turbine Propulsion System are

- *Diffuser*
- *Compressor*
- *Combustion chamber*
- *Turbines*
- *Exhaust nozzle*

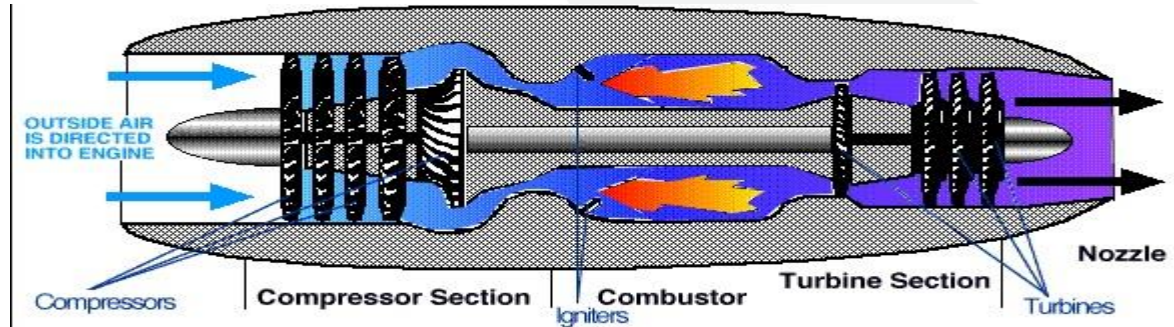


Illustration of components in a Turbojet Engine



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TURBOJET ENGINE

- Most simplest form of gas turbine engine
- Various parts are

- *Diffuser*
- *Compressor*
- *Combustion chamber*
- *Turbines*
- *Tail pipe*
- *Exhaust nozzle*

- All these parts will be housed in a cylindrical shell attached to the wing
- Working is similar to that of a simple gas turbine
- Thrust is derived only by accelerating a huge mass of air through the nozzle.

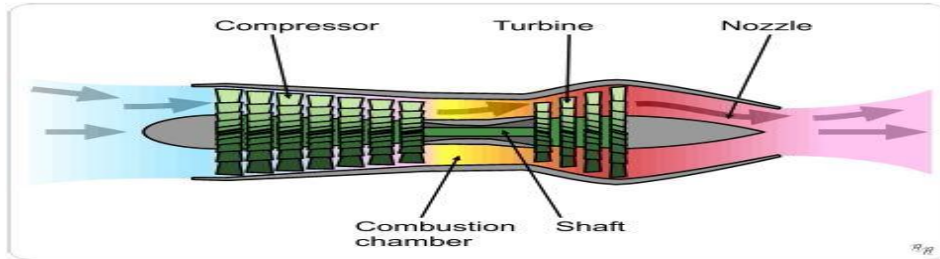
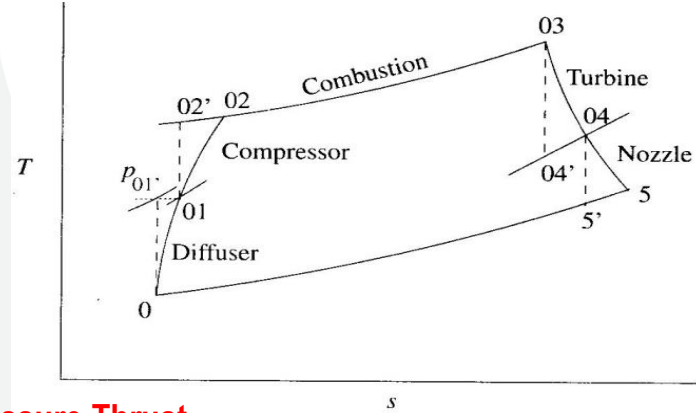
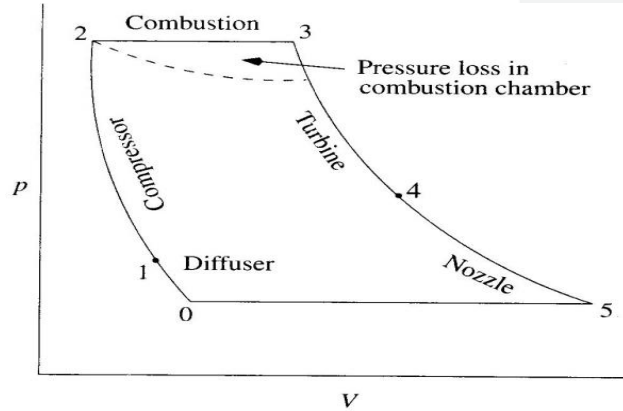


Figure: Schematic of a Turbo jet engine



P-v and T-S diagrams of Turbojet Engine



Total Thrust= Momentum Thrust + Pressure Thrust

$$F_{Total} = F_{momentum} + F_{Pressure}$$

$$F_{moment} = m_j v_j - m_i v_i$$

$$F_{Pressure} = (P_j - P_i) A_j$$

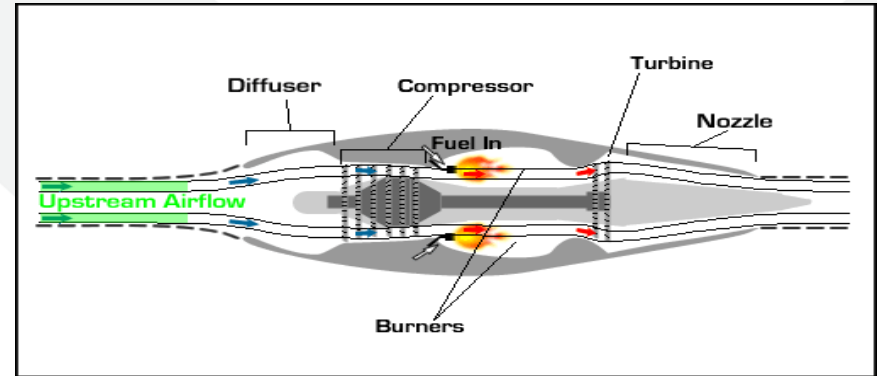
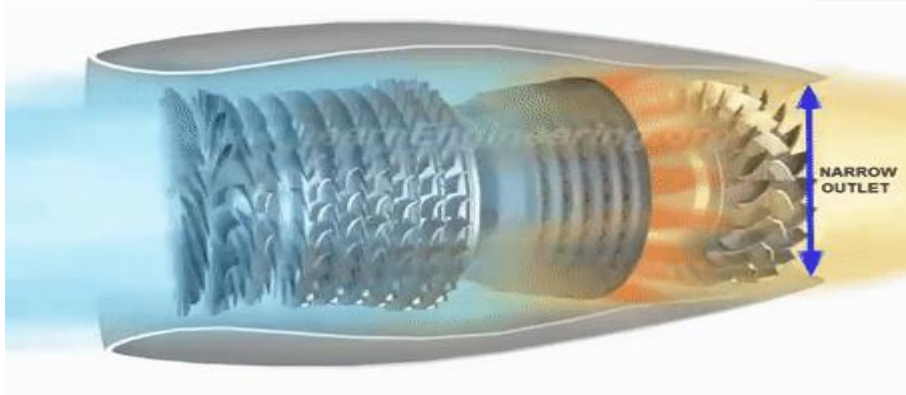
$$F_{Total} = m_j v_j - m_i v_i + (P_j - P_i) A_j$$

$$F_{Total} = \dot{m}_a (V_j - V_i)$$



Characteristics of a Turbojet engine

- Low thrust at low forward speeds
- Relatively high TSFC at low altitudes and speeds
- Lightest specific weight
- Ability to take advantage of highest ram pressure
- Lowest frontal area and almost no ground clearance problem





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Advantages of Turbojet engine

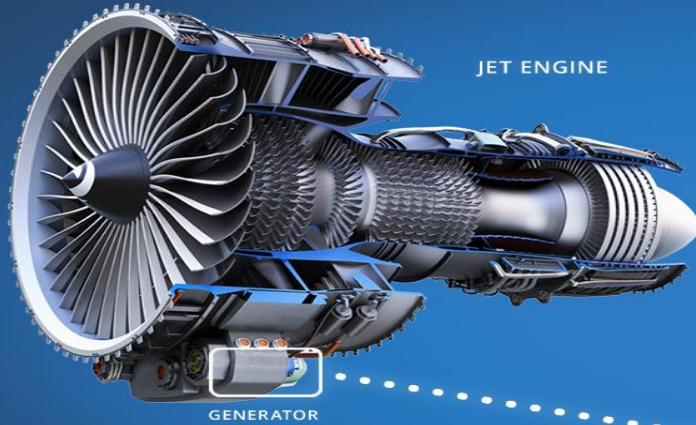
- Ideal for long distance flight at higher speeds and altitudes
- Lower frontal area and shorter landing gear
- Lower weight per unit thrust at design speeds and altitude
- Pressure rise through inlet diffuser is significant
- Reheat can be employed for increased thrust

Disadvantages of Turbojet Engine

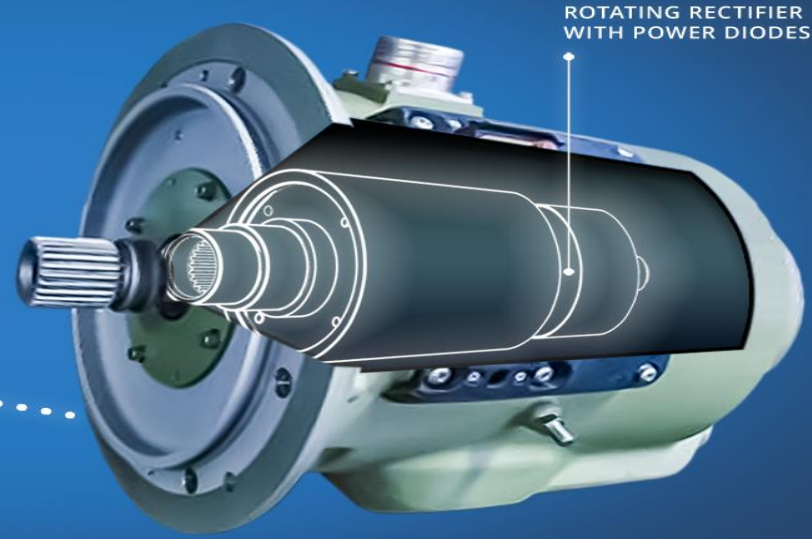
- Take-off roll is longer requiring longer runway
- TSFC is comparatively higher at low speeds and altitudes
- Uneconomical on short distance flights
- Lower thrust and propulsive efficiency at lower speeds.



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JET ENGINE



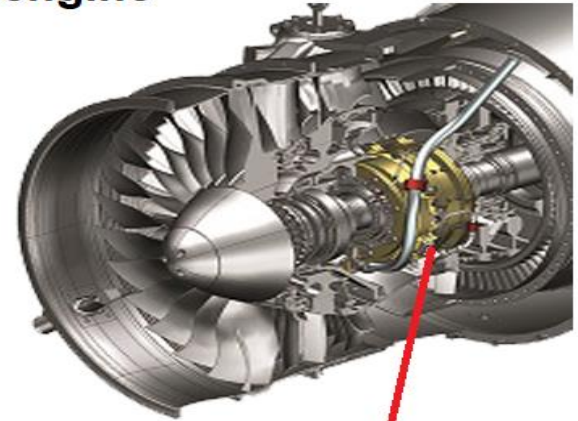
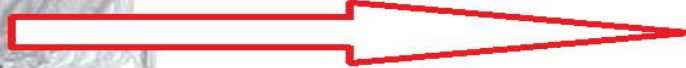
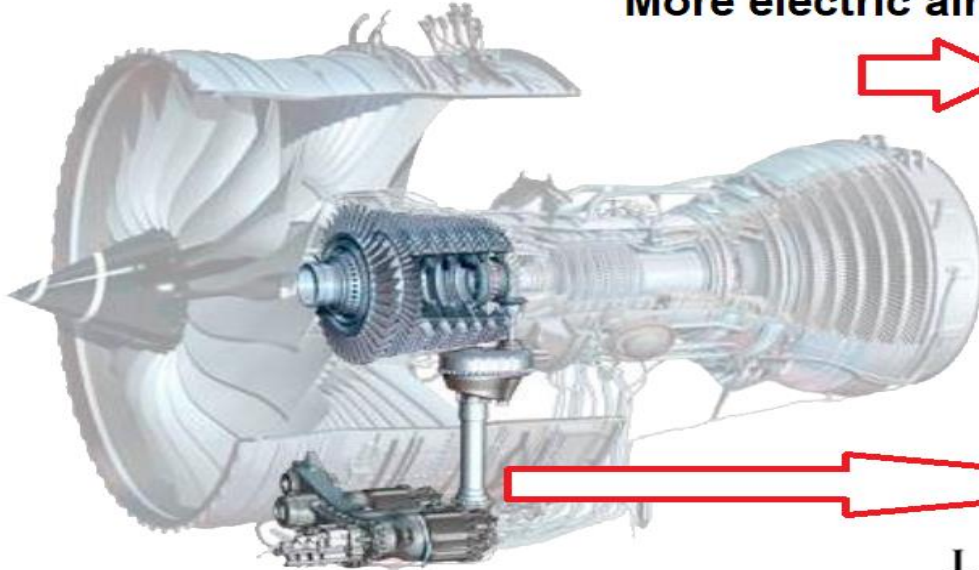
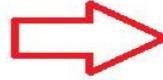
ROTATING RECTIFIER
WITH POWER DIODES



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More electric aircraft/engine



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COMPANY LLC
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Auxiliary Power Unit (APUs)

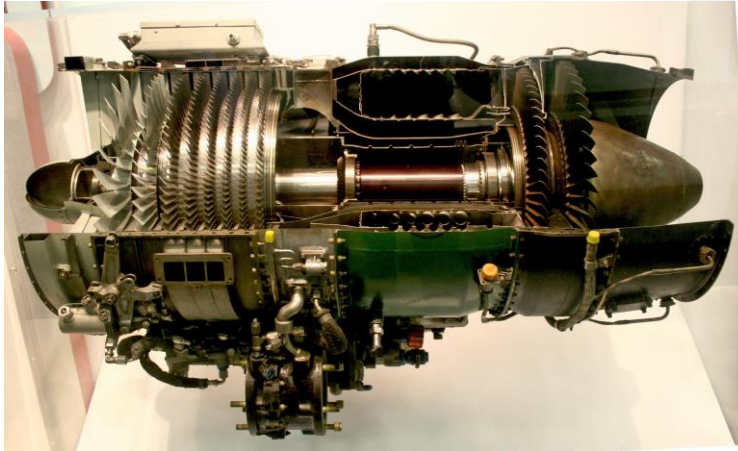
- APU-Provide Electrical Energy
- To power Electrical Systems on Aircraft
- Generally produce-115V AC at 400Hz
- 1 or 3 Phase





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Turbojet Engine example



General Electric J85-GE-17A turbojet engine, circa 1970, which produced a thrust of up to 2,950 lb (13.1 kN).





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TURBOPROP ENGINE

Intermediate of a pure turbojet and a propeller engine

Consists of following parts

- **Propeller**
- **Reduction gear**
- **Compressor**
- **Combustion chamber**
- **Turbine**
- **Nozzle**

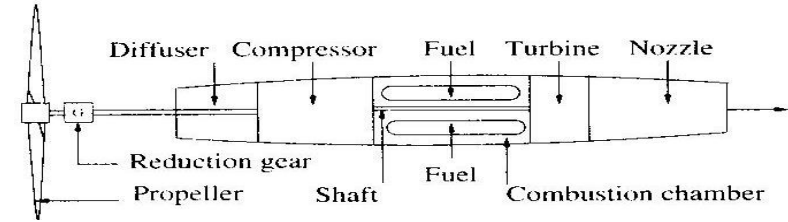
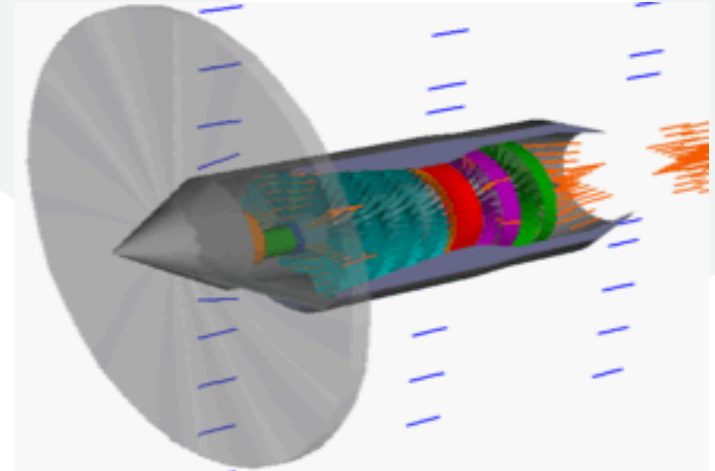
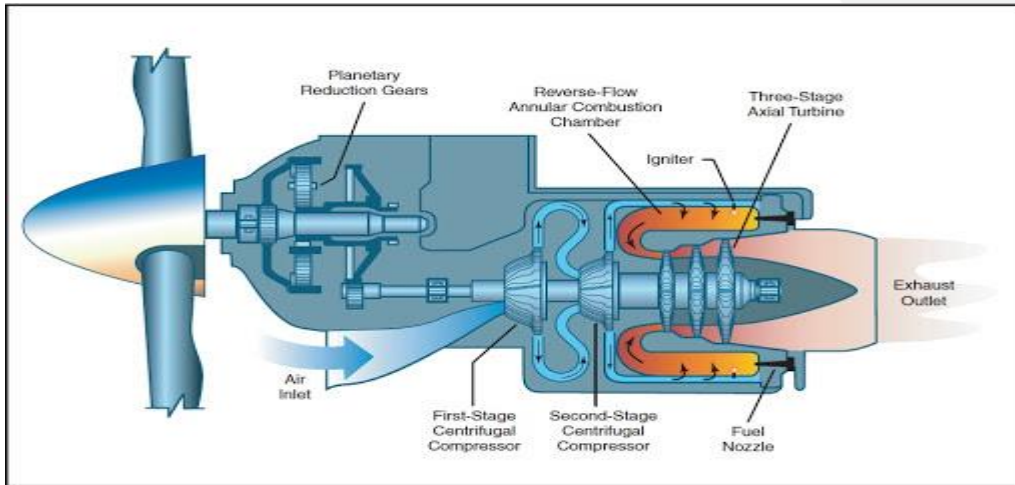


Figure: Schematic of a Turboprop jet engine

➤ Here thrust is generated by propeller as well as the exhaust gases leaving the nozzle.



- **Total Thrust = Nozzle Thrust + Propeller Thrust**

$$F_{Total} = \dot{m}_a (V_2 - V_1) + \dot{m}_a V_1$$





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CHARACTERISTICS OF A TURBOPROP ENGINE

- Very high propulsive efficiency at low airspeeds but with altitude it falls off rapidly
- Most complicated design and heavier than turbojet
- Lowest TSFC
- Large frontal area, Longer landing gear for low wing airplanes
- Highly efficient thrust reversals





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ADVANTAGES OF TURBOPROP ENGINE

- Higher thrust at low speeds
- Take off roll is short
- Propulsive efficiency within operational range is high
- Specific fuel consumption is low
- Thrust reversal is easily achieved



DISADVANTAGES OF TURBOPROP ENGINE

- Application is limited to lower speeds and altitude
- Landing gears have to be longer
- Engine is heavier and complicated
- Usually centrifugal compressors are used which increases the frontal area
- Higher weight per unit thrust



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GE T64 turboprop



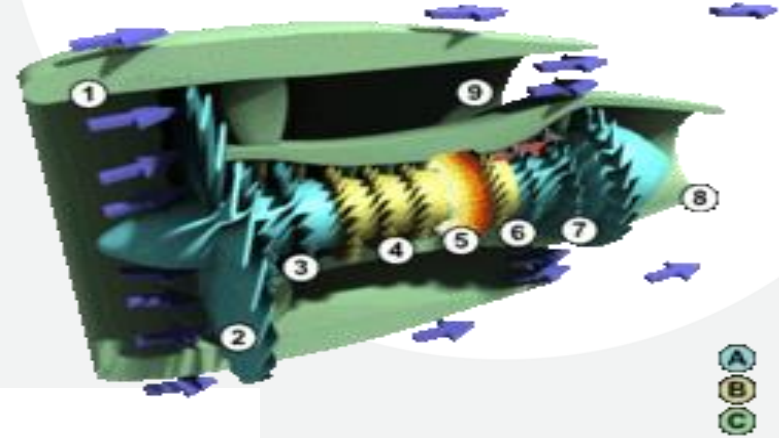


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- Essentially a combination of turbo prop and a turbo jet engine
- It consists of an enclosed duct with a fan mounted either at the front or at the rear end of the engine
- Consists of various parts like

- **Fan**
- **Compressor**
- **Burner**
- **Turbine**
- **Nozzle**

TURBOFAN ENGINE



Turbofan Engine

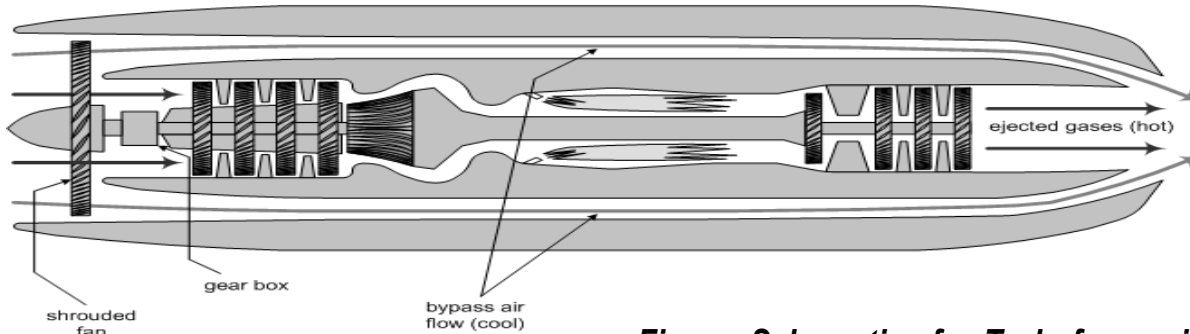


Figure: Schematic of a Turbofan engine



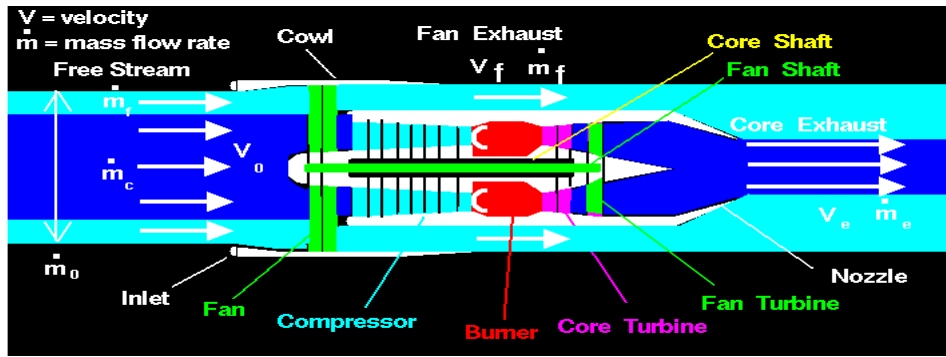
Thrust Equation for a Turbofan Engine

- **Total Thrust = Fan Thrust+ Nozzle Thrust**

$$F_{Total} = (\dot{m}_f V_f - \dot{m}_f V_0) + (\dot{m}_e V_e - \dot{m}_c V_0)$$

$$F_{Total} = bpr \dot{m}_c V_f + (\dot{m}_e V_e - \dot{m}_c V_0)$$

$$bpr = \frac{\dot{m}_f}{\dot{m}_c}$$





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CHARACTERISTICS OF TURBOFAN ENGINE

- Increased thrust at forward speeds
- Can fly beyond Mach 1
- Weight falls between turbojet and Turboprop
- Ground clearances are better than turboprop but not as good as turbojet
- TSFC and specific weight falls between Turbojet and turboprop
- Lesser noise level
- Performance on a hot day is superior to the turbojet
- Separate thrust reversers are required if Primary and fan air exit separately





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Short take off roll

- Comparatively quieter engine
- Weight per unit thrust is lower than turboprop engine
- Thrust is higher than turbojet engine

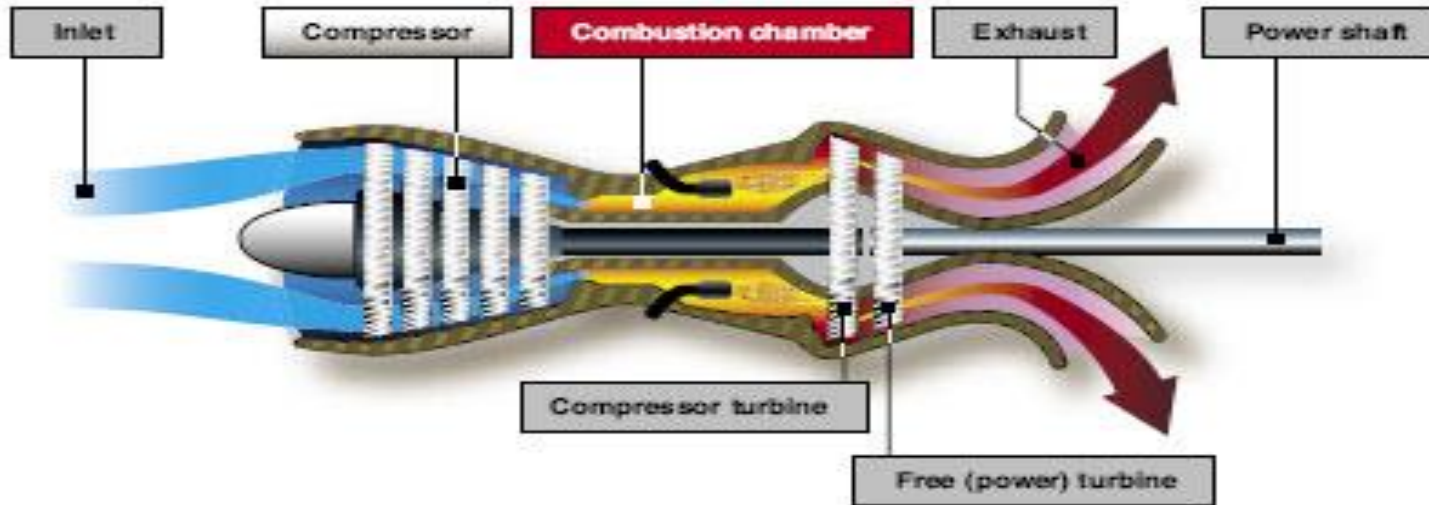
ADVANTAGES OF TURBOFAN ENGINE

DISADVANTAGES OF TURBOFAN ENGINE

- Increased frontal area and drag
- Separate thrust reversers are required for hot and cold jets
- TSFC is higher than turboprop
- Engine is heavier and complicated than turbojet engine
- Lower speed limit than turbojet engine $M < 1$



TURBOSHAFT ENGINES

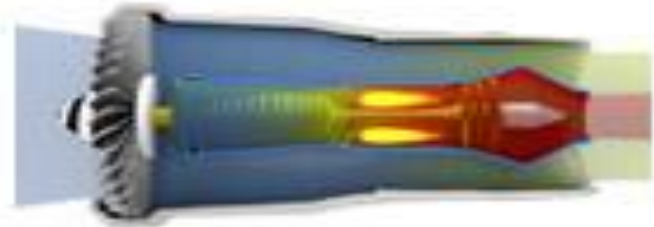




VARIOUS CONFIGURATIONS OF GAS TURBINE ENGINES



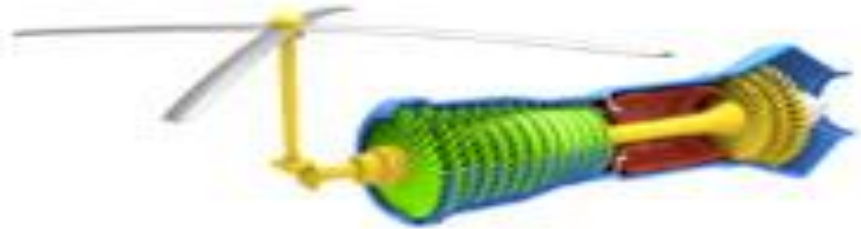
Turbojet



Turbofan



Turboprop



Turboshaft



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- Simplest form of air breathing engines
- Compressors and turbines are absent
- Speed ranges between Mach 3-6
- Rely on RAM effect
- Pressure ratio of around 30 can be easily achieved

WORKING ENVIRONMENT FOR COMPRESSOR

Subsonic velocity ($M=0.3$ to $M=0.5$)

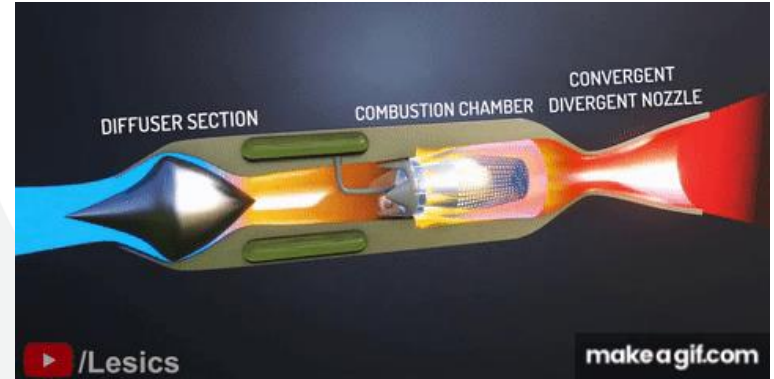
Laminar flow

Minimum residence time for compression

ADVERSE CONDITIONS EXPERIENCED BY A COMPRESSOR

- Shock formation at blade tips
- High drag on the blades
- Turbulent flow
- Internal shock reflections
- High stage losses
- Shock induced vibrations
- Maximum Pressure ratio $r=20$ to 25

RAMJET ENGINES

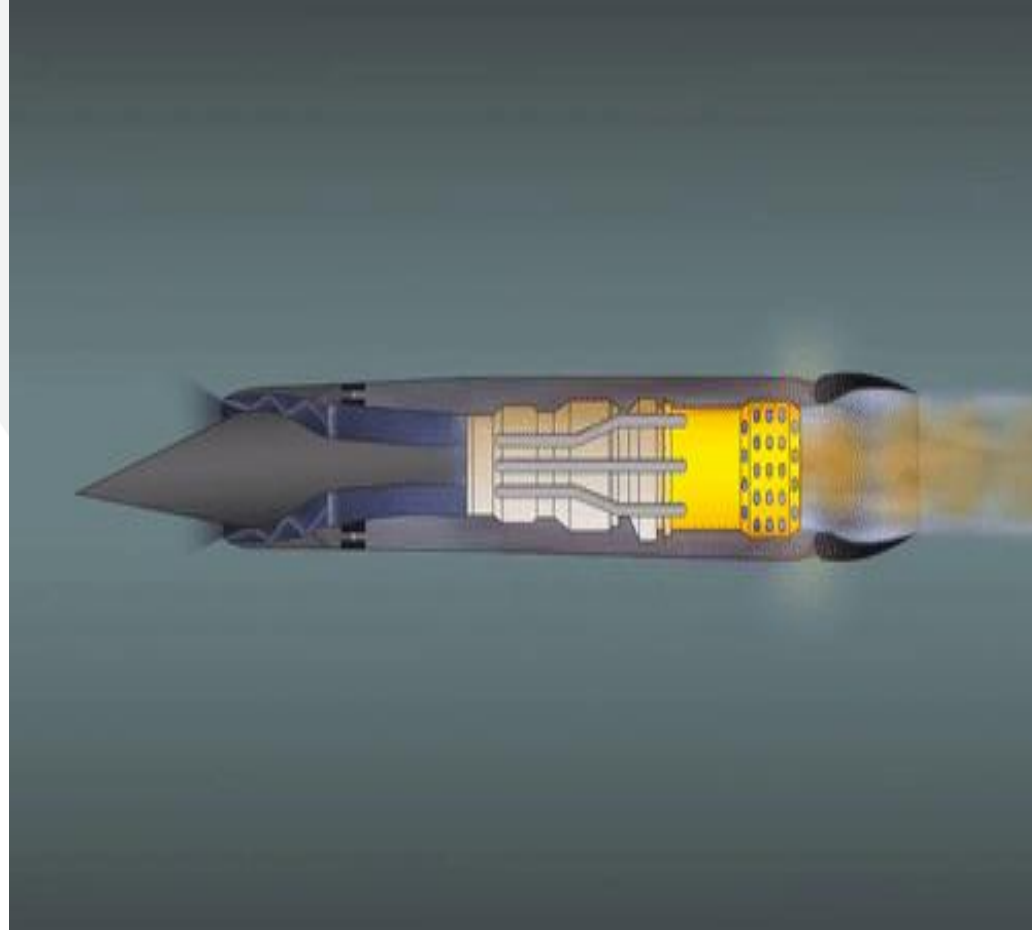
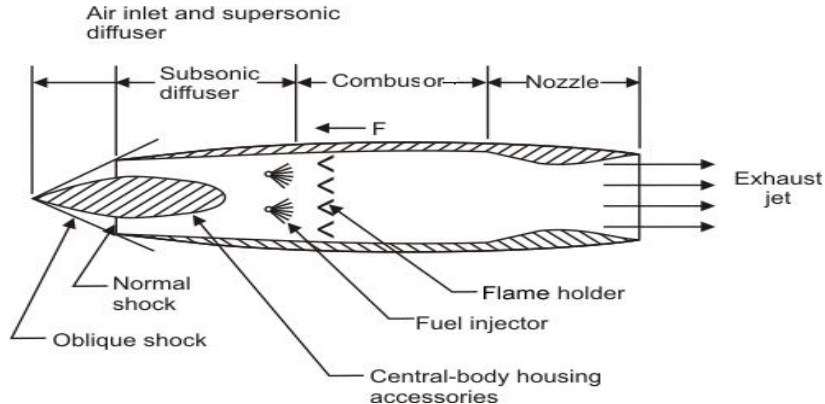




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WORKING OF A RAMJET ENGINE

- Relies on RAM Effect to produce thrust
- Parts are
 - *Supersonic Diffuser*
 - *Subsonic Diffuser*
 - *Combustion Chamber*
 - *Exhaust Nozzle*
- Generation of Shock waves
 - *Oblique Shock waves*
 - *Normal Shock waves*



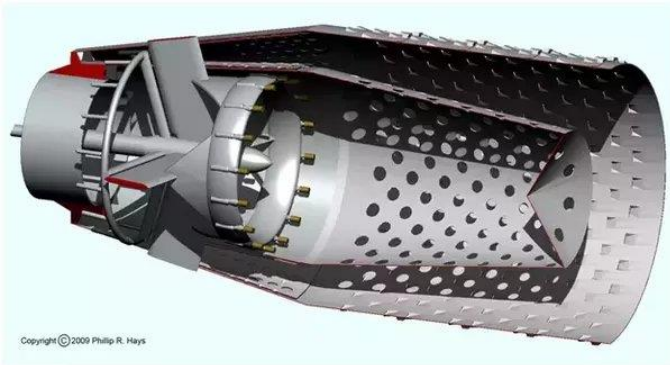


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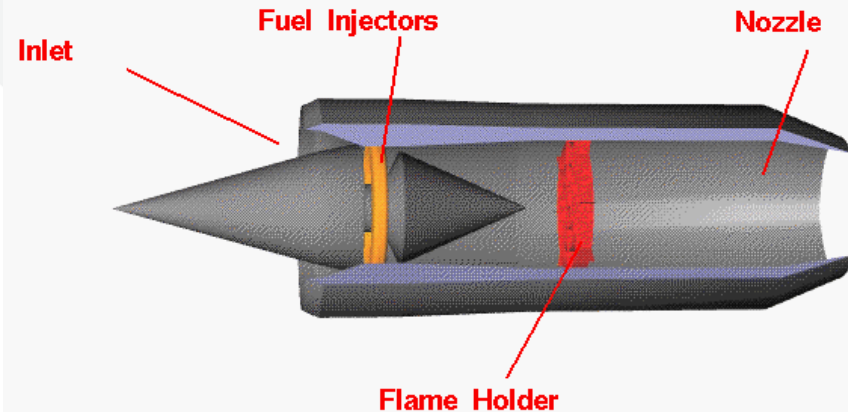




- Fuel-hydrocarbon fuel-kerosene, JP-7, JP-10
 - High energy content, stability, and availability
- Operate at stoichiometric fuel:air ratios
- Combustor exit stagnation temperature 2,130 °c-kerosene
- Mach 3 (3,700 km/h) to mach 6 (7,400 km/h)



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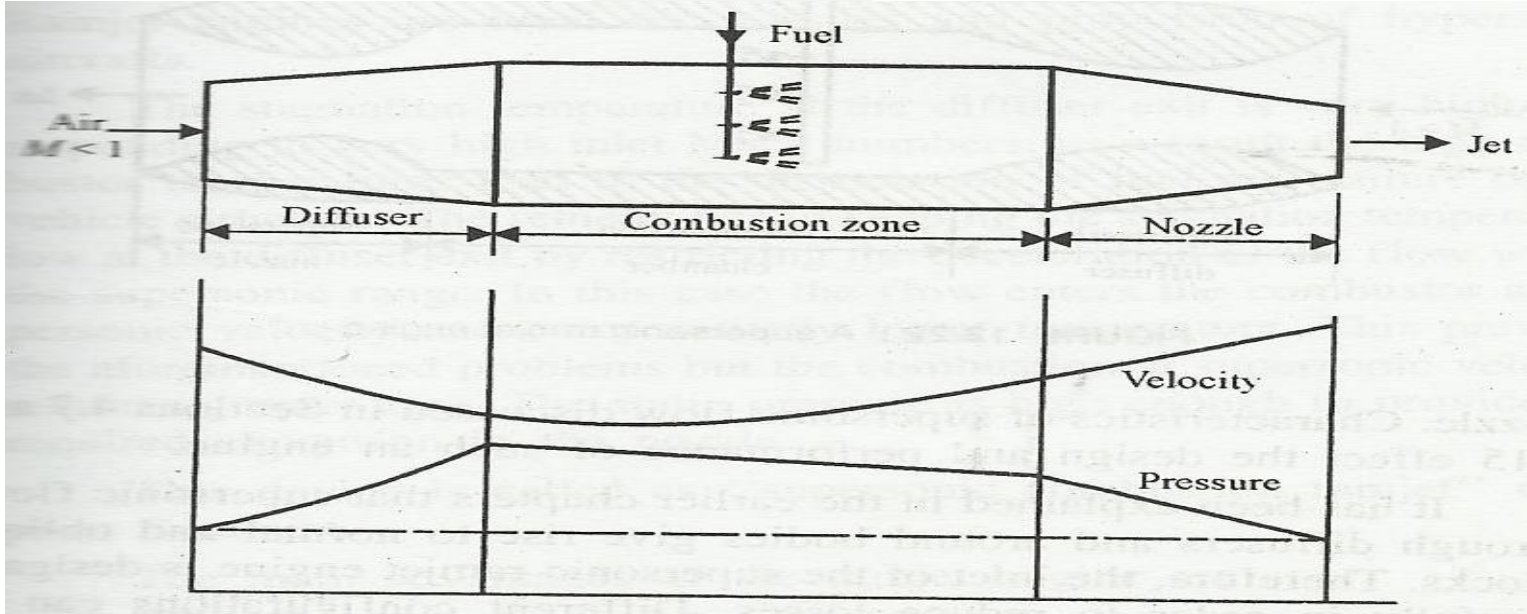


Fig : Pressure & Velocity variations across the Ramjet Engine



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- External launch vehicle
- From an airplane in flight
- External boosters
- Solid-propellant rockets





ADVANTAGES OF RAMJET ENGINES

- High temperatures can be employed
- In the absence of rotating machinery its construction is very simple and cheap
- It can operate efficiently at high supersonic Mach numbers
- It is not very sensitive to the quality of fuel
- It provides high thrust per unit weight and frontal area

DISADVANTAGES OF RAMJET ENGINES

- It requires a specialized launching device to take off
- It is not suitable for subsonic speeds
- It has low thermal efficiency and high TSFC
- Its maximum operating altitude is limited



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CHARACTERISTICS OF RAMJET ENGINES

- Construction is simple.
- Possibility of Mass production-Especially for Missiles
- Relatively low cost
- Wide range of fuels can be employed-Solid, Liquid etc
- High SFC at low and moderate speeds
- SFC decreases with Flight Speed
- Suitable for propelling supersonic missiles



Applications of RAMJET Engine

- SR-71 Blackbird- Turbo-RAMjet Engine
- BrahMos Missiles



Inlet of the J58 engine on the SR-71 Blackbird aircraft features a prominent inlet spike, which can be seen in the photo below (*courtesy of Charles Norren*). The role of this inlet spike is to compress and redirect the incoming air into the engine.



An interesting feature of this inlet spike was its ability to retract at high speeds to position the external oblique shock and the internal shocks correctly in order to direct the appropriate amount of the airflow into the engine. The overall flow geometry of the inlet is illustrated below.



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- BrahMos Missile





SCRAMJET (SUPER SONIC COMBUSTION RAMJET)

- Operate in the hypersonic flight regime
- Same principles as ramjets
- Does not decelerate the flow to subsonic velocities
- Velocity of air is supersonic
- Accelerated in nozzle
- The **scramjet** is composed of three basic components:
 - Converging inlet
 - Combustor
 - Diverging nozzle

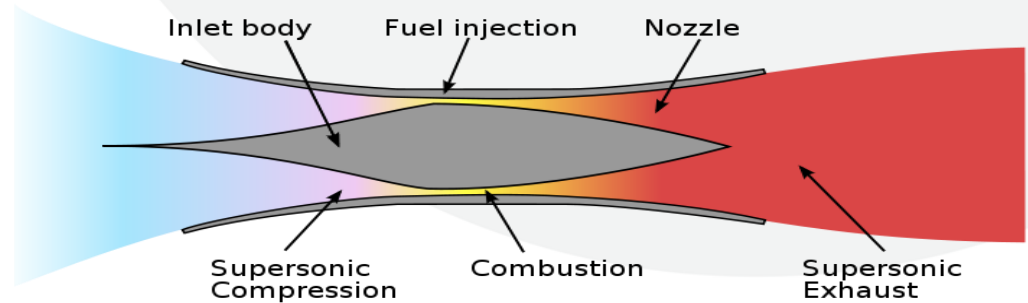
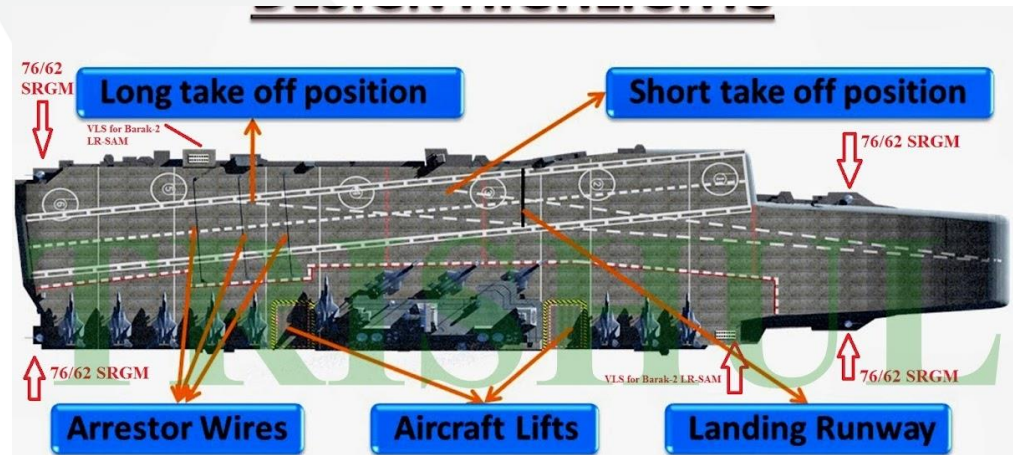


Fig: Schematic Diagram of a Scramjet Engine



THRUST AUGMENTATION

- Turbojet engine has only BASIC THRUST
- Cannot perform specialized missions
 - *Take-off from short runway*
 - *Take-off on a hot summer day*
 - *During Dog fighting*
 - *Slam Acceleration*
 - *Instantaneous Vertical Climb*
 - *Supersonic Cruise*





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Techniques to improve Basic Thrust

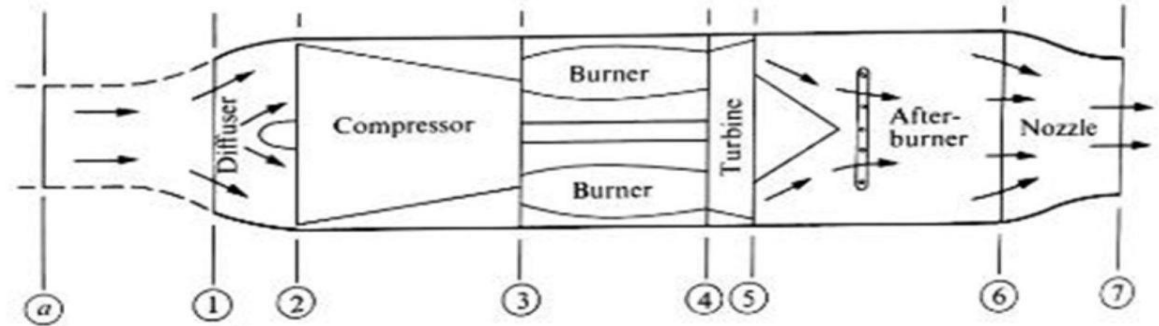
- Increase thrust of the engine
 - Increase Size, but with penalties
- Three techniques
 - *Afterburning Mainly Military Aircrafts but also CONCORDE*
 - *Water or Water-Alcohol injection (Water + Methanol)*
 - *Bleed Burn cycle*





AFTERBURNING

- Auxiliary Burner downstream
- Burn additional fuel
- Presence of Variable Nozzle
- 50 % additional thrust
- Excess fuel consumption

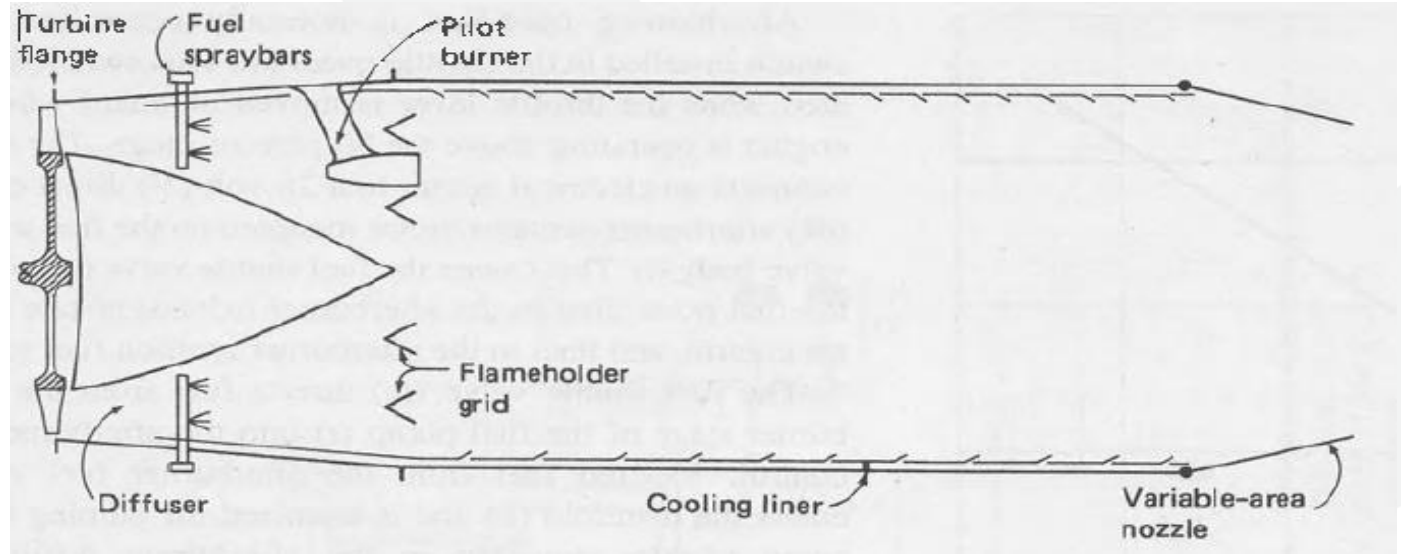


Considering the schematic diagram of the ideal turbojet provided, please describe the process going from one station to the next as if you were sketching a T-s diagram. Consider all of the stations shown in the diagram (a – 7). Your answer does not need to be paragraphs long but the process from one location to another does need to be accurately described in the context of a T-s diagram.

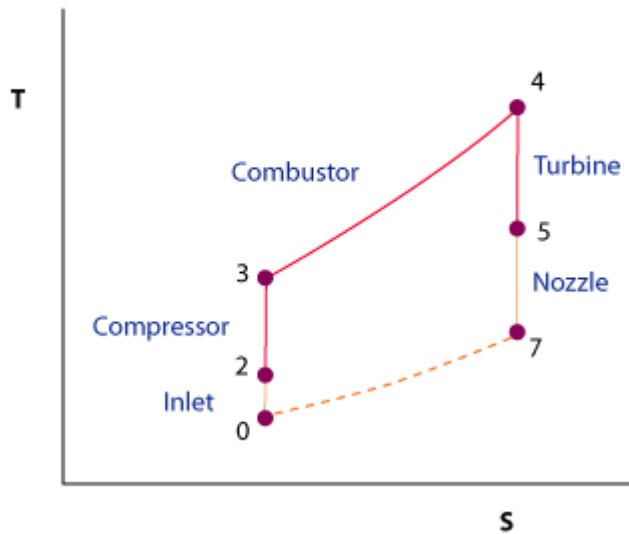


Basic components of afterburner

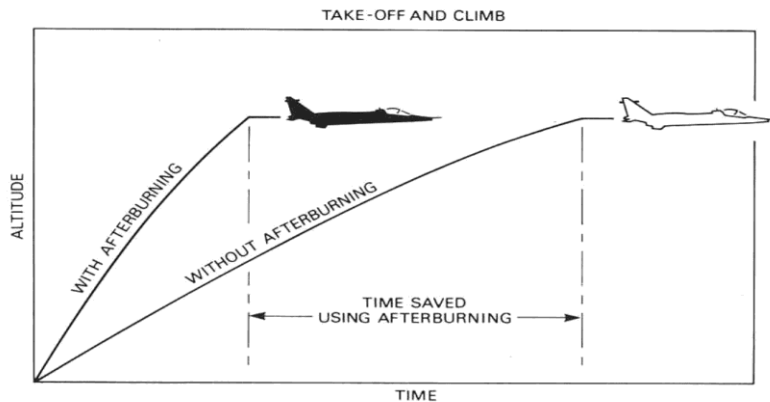
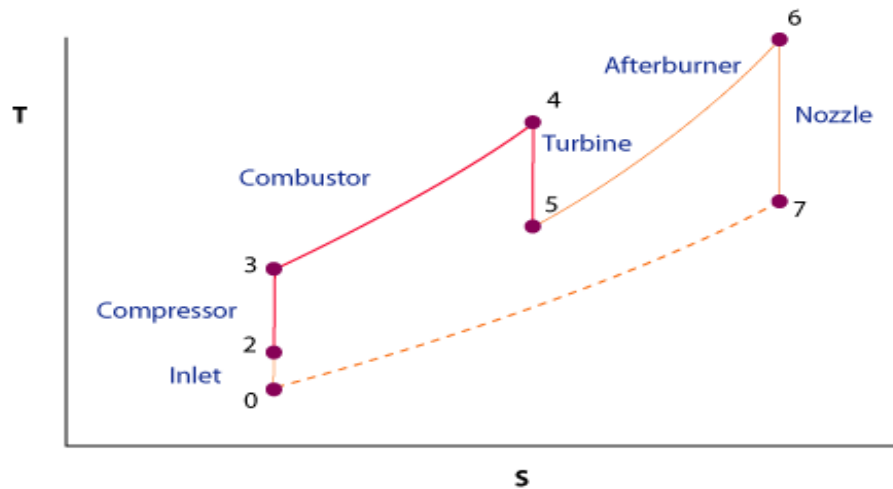
- Diffuser
- Spray Bars
- Torch Igniters
- Flame Holder
- Variable Area Nozzle
- Screech Liners



Turbojet Cycle

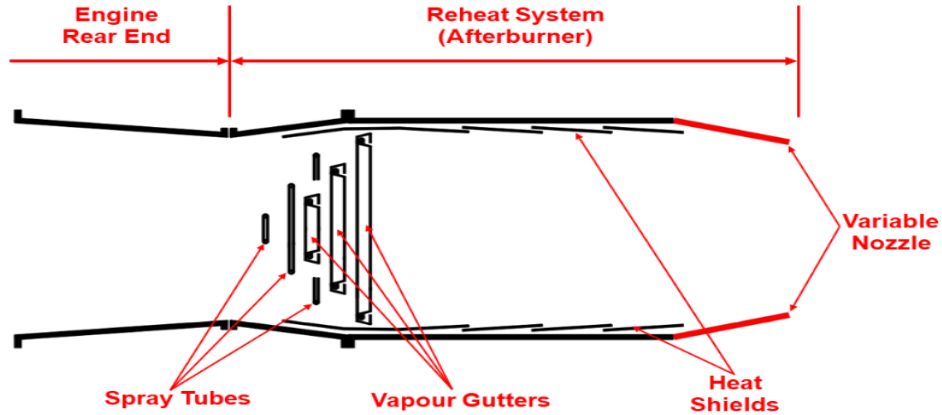


Turbojet-Afterburner Cycle



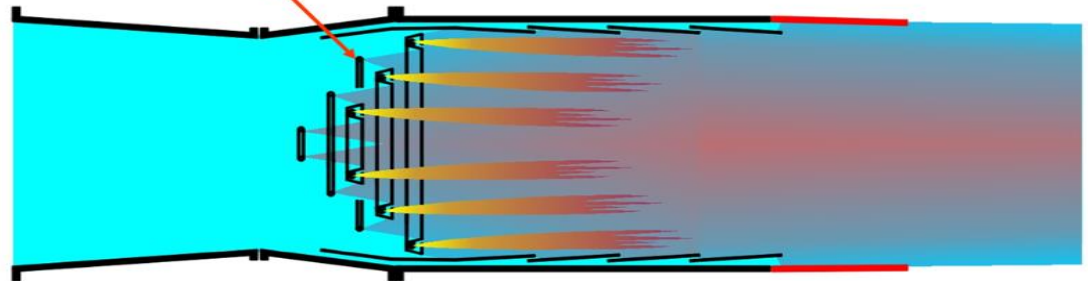


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Then Spray Tubes add more
fuel to combustion process

And Variable
Nozzle 'Opens'

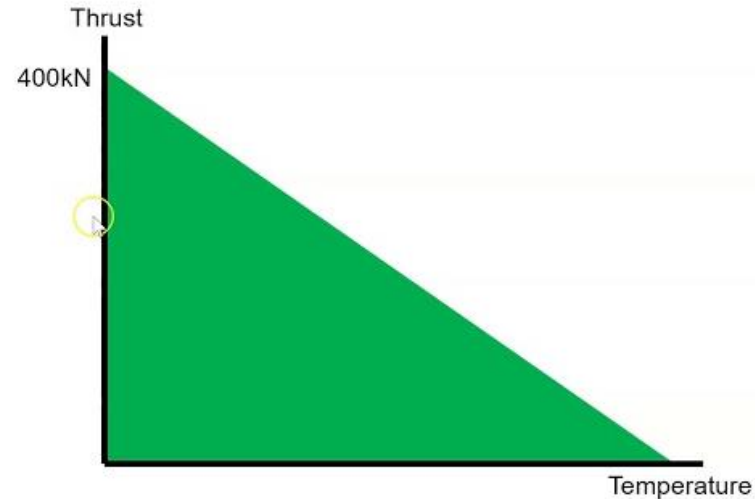




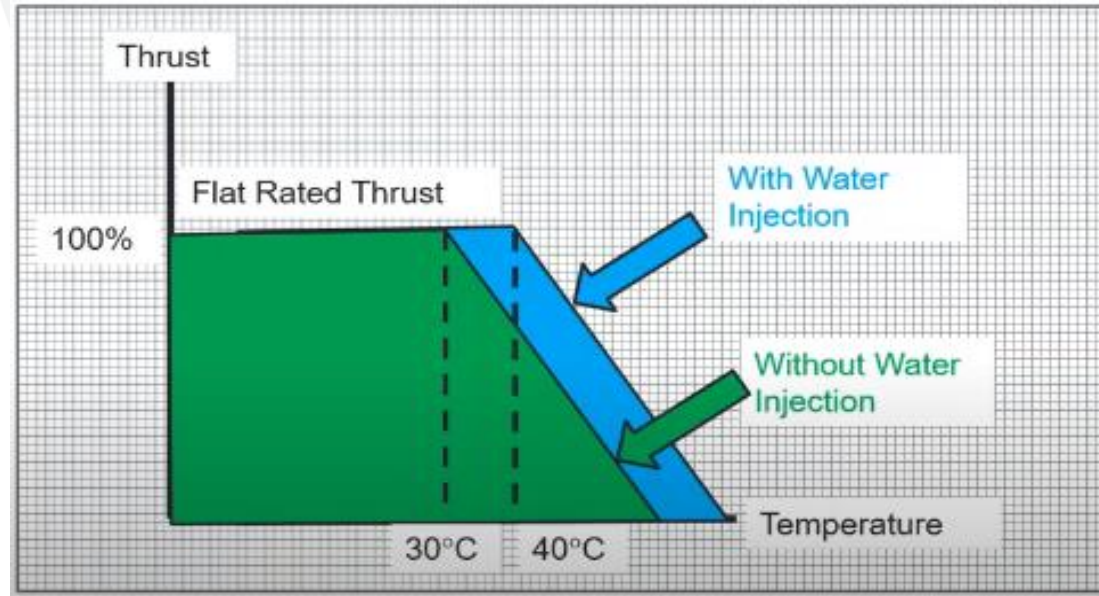
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- Simplest technique
- Used when thrust loss due to pressure
- 30% extra thrust is obtained

Flat Rated Engine



WATER INJECTION





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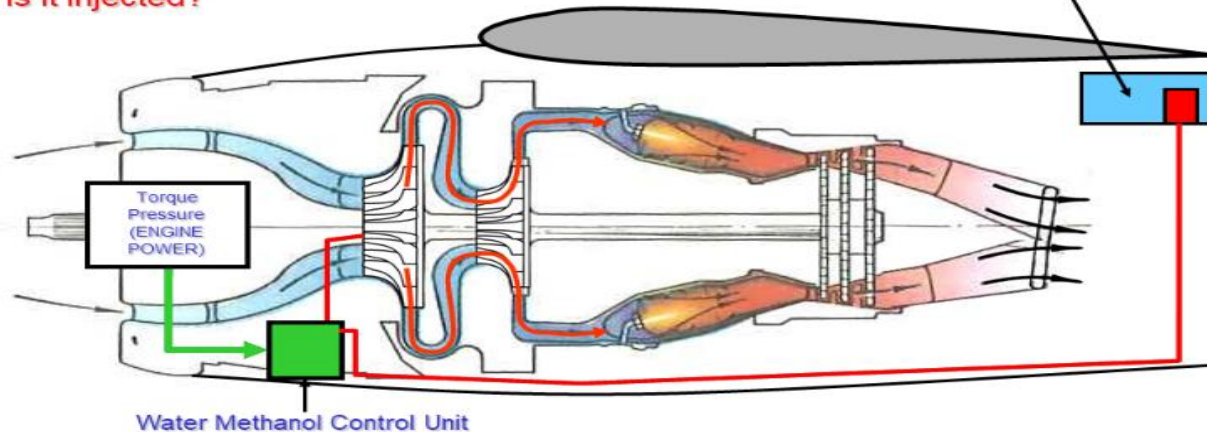
- Simplest technique
- Used when thrust loss due to pressure
- 30% extra thrust is obtained
- Provides cooling effect in the compressor
- Increases pressure ratio without increasing the turbine work

Water Methanol Injection

How is it injected?

Dart Engine Series

Water Methanol
Tank and Pump

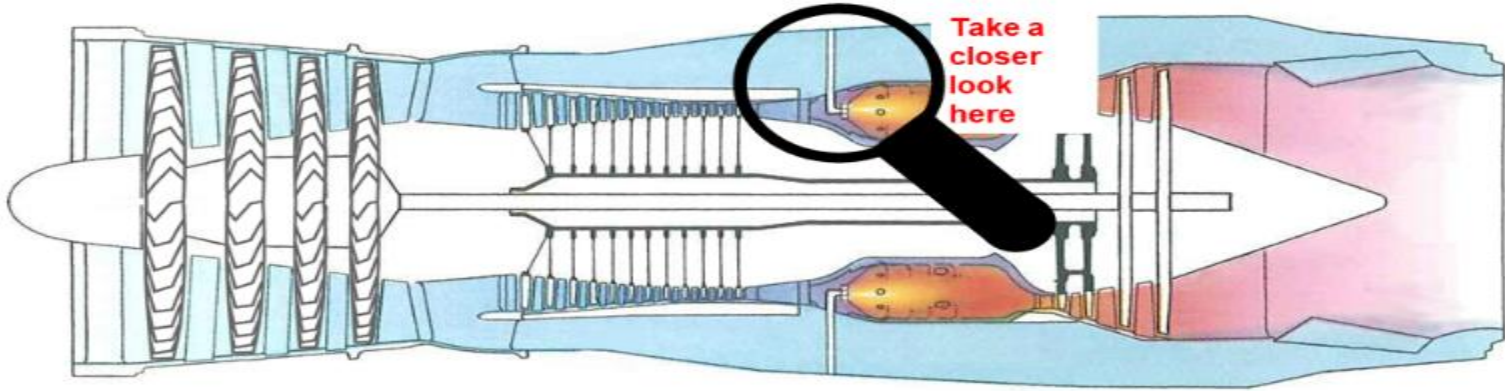
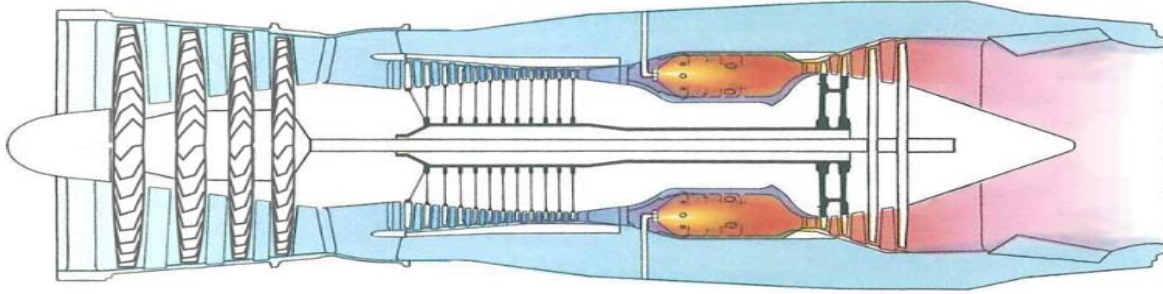




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Water Injection

Rolls-Royce Spey Series

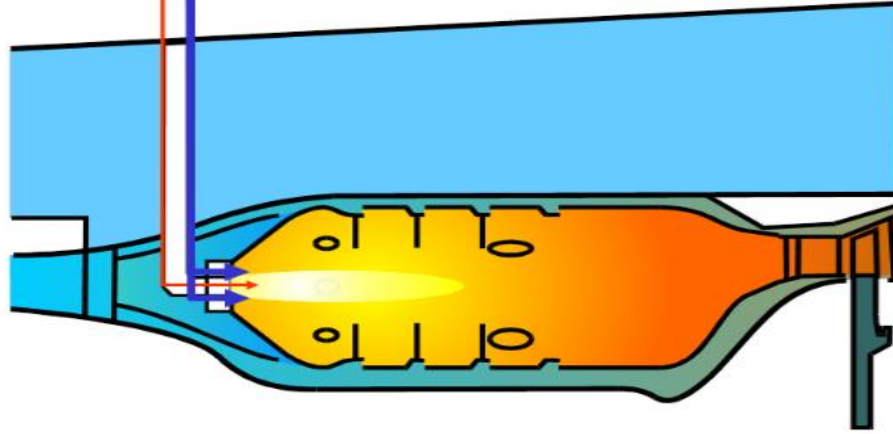




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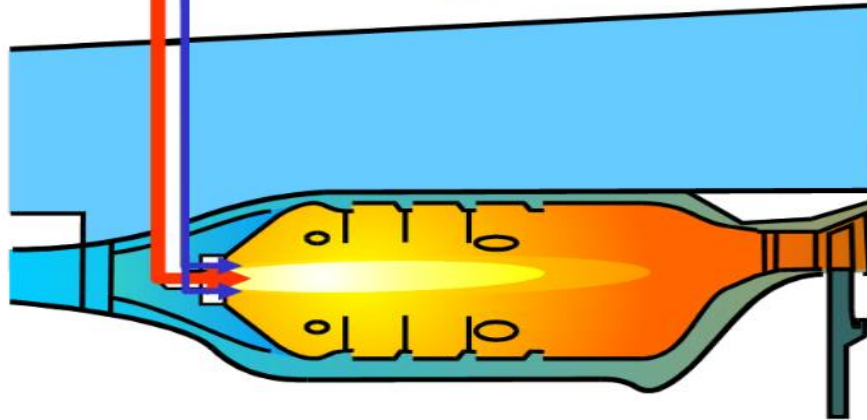
Primary
Fuel flow

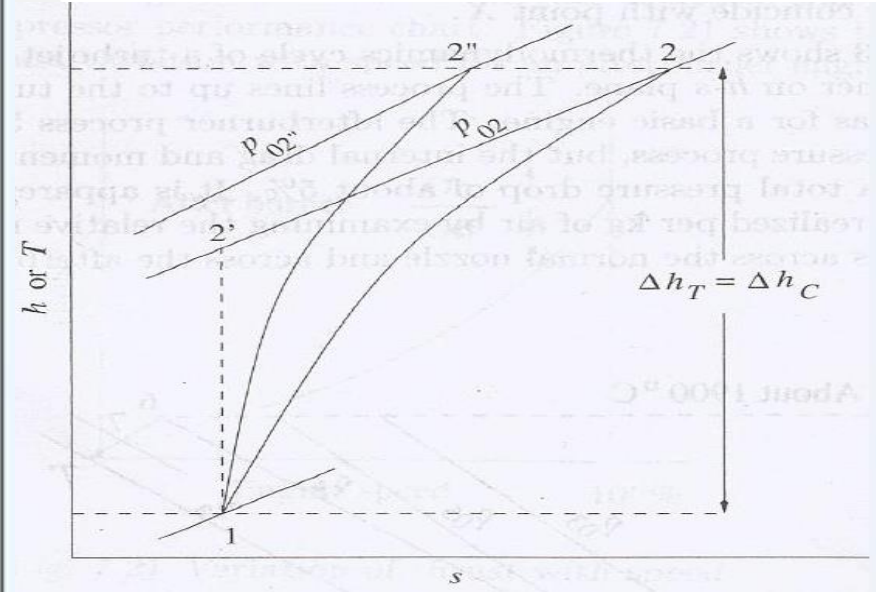
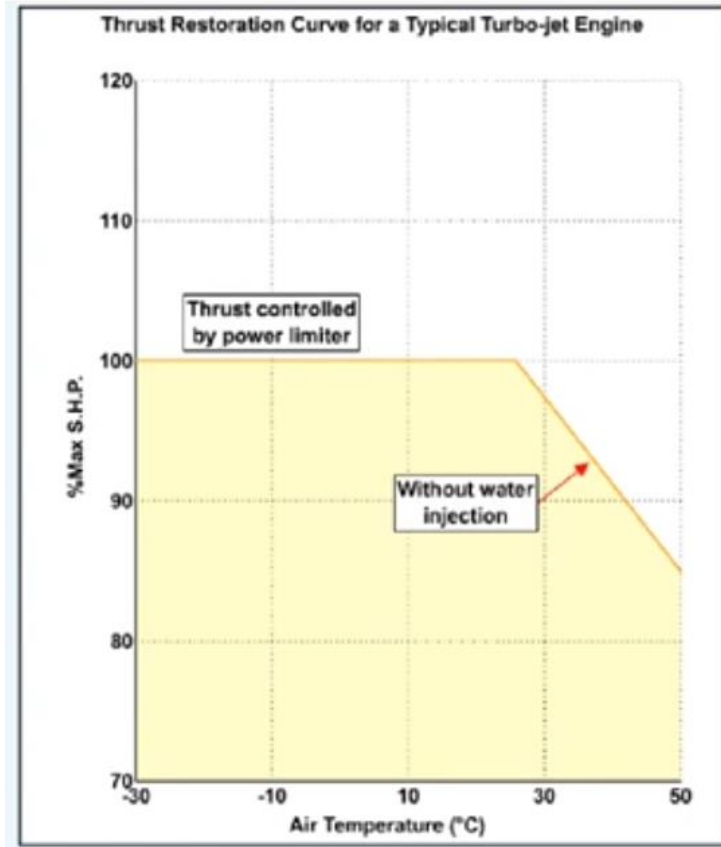
Water
flow



Increased
Primary
Fuel flow

Water
flow







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BLEED BURN CYCLE

- Produces maximum thrust
- Burns additional fuel in secondary combustor
- Large Fuel consumption

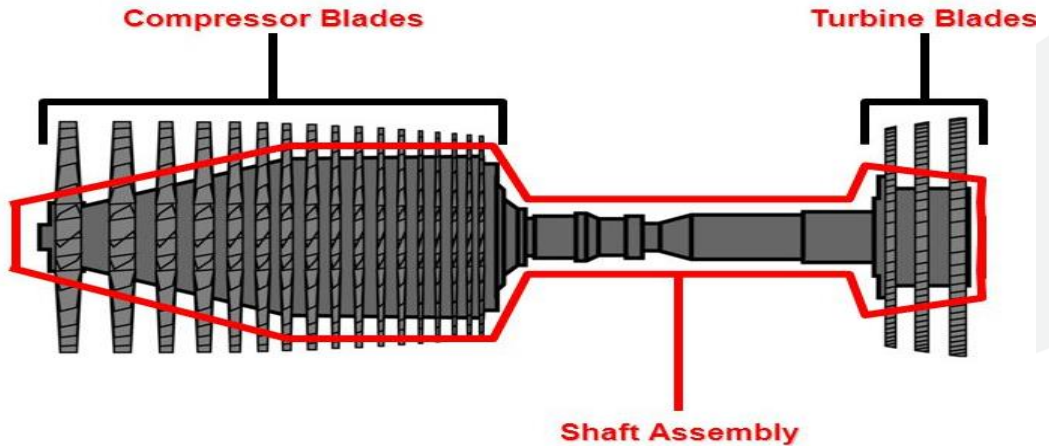


SPOOLING OF JET ENGINES

- A compressor and a turbine connected through a common shaft -SPOOL
- Greatest Innovation in jet engines
- Invented by Pratt & Whitney
- Significantly improves the performance

TYPES

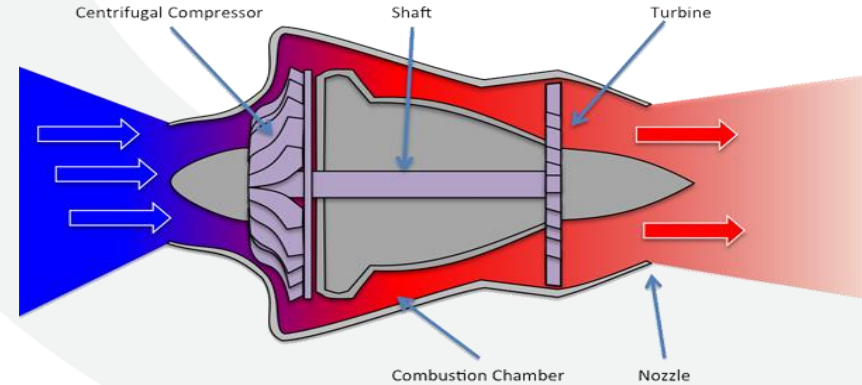
1. Single Spool Jet Engines
2. Twin Spool Jet Engines
3. Three Spool Jet Engines





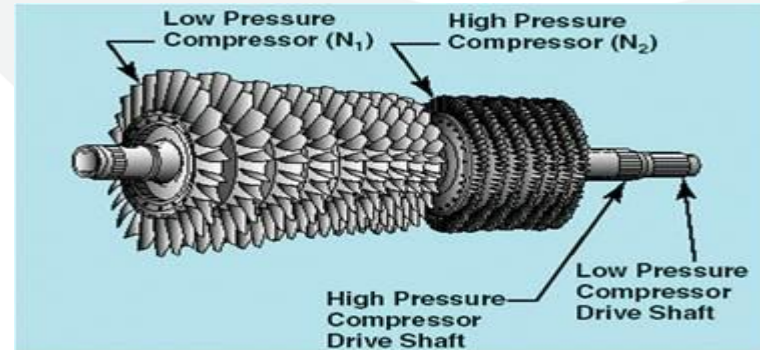
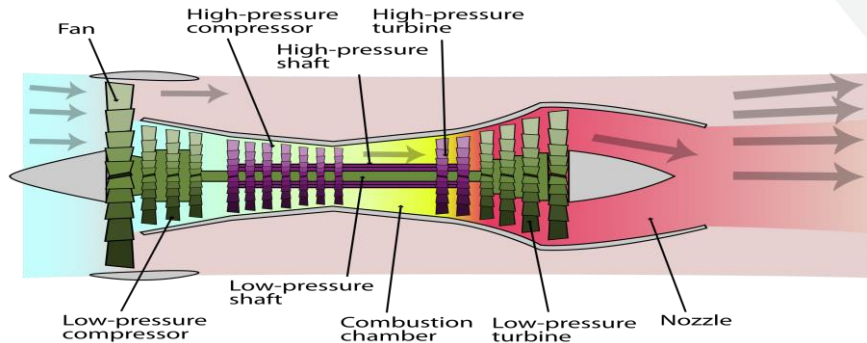
SINGLE SPOOL JET ENGINES

- One compressor and a turbine on a single shaft
- Includes Fan and a propeller-Turboprop and Turbofan
- Thrust depends on Pressure ratio and Turbine Inlet Temperature (T_3)
- Increasing the Pressure ratio increases
 - Number of stages
 - Overall Length
 - Flow losses
 - Stresses on the blades and discs
 - Increasing stresses require thick blades and casing which increases the weight, fuel, drag etc
- $N_1=N_2$
- Speed of compressor is dependent on turbine



TWIN SPOOL JET ENGINES

- Compressor/turbine units are split into two independent unit
- Two compressor and two turbines
 - Low pressure and High pressure compressor
 - Low pressure and High pressure turbine
- Connected through two concentric shafts-Hollow and solid
- Reduces the work of compression (compressor) or increases the work of expansion (turbine)
- Independently run at different optimum speeds

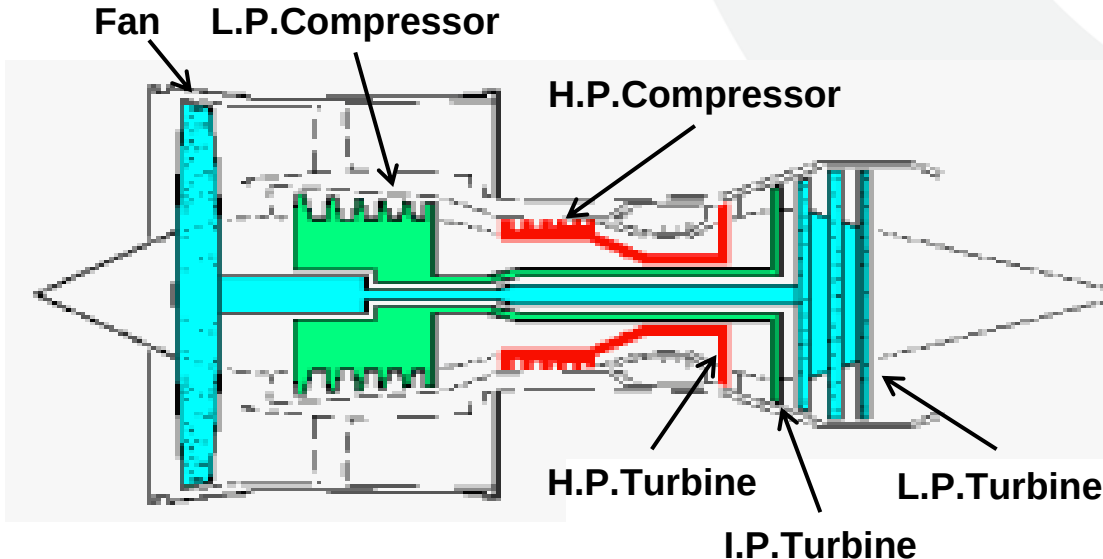




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THREE SPOOL JET ENGINES

- Geometrically complex
- Low pressure, intermediate and high pressure units
- Fan connected to low pressure turbine
- Low pressure compressor to intermediate turbine
- High pressure compressor to high pressure turbine



Spool-1 (Blue)
Fan & L.P.Turbine

Spool-2 (Green)
L.P.Compressor &
I.P.Turbine

Spool-3 (Red)
H.P.Compressor &
H.P.Turbine



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THREE SPOOL JET ENGINES

ADVANTAGES

1. More Thrust during take-off
2. Reduces stress and load on the engine
3. Increased fuel economy

DISADVANTAGES

1. Increased complexities
2. Prone to vibrations
3. Internal cooling is complex and requires more cooling
4. Emit more smoke
5. Makes the engine heavier
6. More rotational inertia, hence slow acceleration
7. Lubrication and oil distribution is expensive and complex
8. Dissipate more heat



- GE-NX Turbofan Engines-Specifications

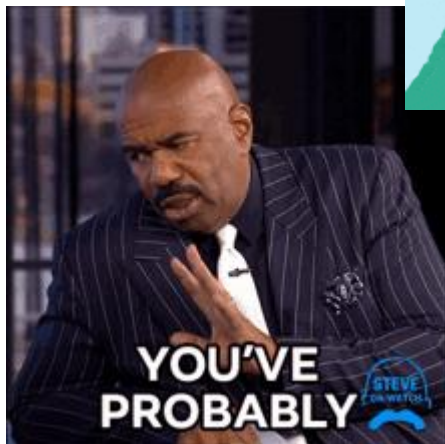


Title	Sym- bol	Value	Unit
Width	-	3,533.1	mm
Height	-	3,484.9	mm
Overall Length	-	4,950.5	mm
Intake/Fan Diameter	D_f	2.82	m
Max. Take-off Thrust (5 min.)	T_{max-TO}	321.735	kN
Fan speed at Take-off Thrust (5 min.)	N_{f-TO}	2,401	rpm
Max. continuous thrust	T_{max-C}	295.8	KN
Fan speed (max. continuous thrust)	N_{f-C}	2,319	rpm
Pressure Ratio of High-Pressure Compressor	π_{HPC}	23	-
Overall Pressure Ratio (at max. power)	OPR_{Max}	43.5	-
Fuel flow rate(Take-off)	$\dot{m}_{f-TO}$	2.037	rpm
Fuel flow rate (Cruise)	$\dot{m}_{f-Cruise}$	2.494	m/s
Bypass ratio	β	9.1	-
Total air mass flow rate (Take-off)	$\dot{m}_{a-TO}$	1155.43	kg/s
Bypass air flow rate	$\dot{m}_{a-BP}$	1041.03	kg/s
Core air flow rate	$\dot{m}_a$	114.399	kg/s
Fuel heating value	Q_R	42555.066	kJ/kg

→ **FEEDBACK**

FEEDBACK

**GOOD
JOB!**



Feedback !?

